UNIT-1

Electrical and Electronics Engineering [205]

INTRODUCTION

Electricity is a form of energy. It can be generated in huge quantity at a suitable location. It can be transmitted over long distance very economically. Electricity is applicable in almost all activities of our life such as lighting, heating, cooking, transportation, agriculture, Industrial, Commercial and most of domestic applications.

UNIT-I D.C. NETWORKS.

ELECTRIC CHARGE

Each and every thing in this universe is made up of molecules which can be further divided into atoms. The atoms are made up of electrons, protons and neutrons. The electrons and protons posses -ve and +ve charge (1.602×10^{-19} coulombs). These charges are natural. The quantity of +ve charge on proton is equal to the quantity of -ve charge on electron. Every atom has equal number of protons and electrons. Hence net charge on an atom is zero. Neutrons have no charge. Hence all atoms and material behaves as neutral.

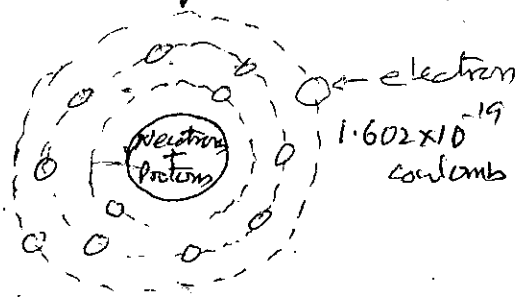


Fig. 1

CTRIC CURRENT

The protons and neutrons are tightly bound its nucleous and can not move out of the n. whereas, electrons of an atom encircle its leous in different orbits. The electrons of an in its outermost orbit can move from one to another atom easily. when one or more tron of an atom left its parent atom the atom looses its -ve charge and in condition the atom becomes +vely charge is called an ion. These free electrons & are also called charge carriers. These ge carriers move randomly in different ctions about their mean position and hence displacement is zero. when an electric field ectric pressure is applied across any conductor electrons drift in a particular direction and drift of electrons makes electric current.

mple :- There is a drift of 5×10^{19} electrons per second across a cross section of a conductor. Find the current in the conductor.

$$= \quad I = \frac{dQ}{dt}$$

$$= \frac{5\times10^{19}\times1.6\times10^{-19}}{1} = 5\times1.6 = 8 \text{ Ampere}$$

Ans.

where I is current in Ampere.
Q is charge in coulomb.
t is time in second.

So the drift or movement of electrons in one direction in a conductor is known as electric current or it can also be said that flow of current in a conductor takes place by the movement of electrons in a particular direction. The direction of flow of current is opposite to the direction of drift of electrons. current can also be defined as charge in motion.

Since the charge on an electron is 1.602×10^{-19} coulomb. Therefore one ampere of current takes place when $\frac{1}{1.602 \times 10^{-19}} = 6.24 \times 10^{18}$ electrons flow per second through the cross section of a conductor.

VOLTAGE

The free electrons of an atom moves randomly in different directions. To move these free electrons in a particular direction to produce current, an electric force or electric pressure is required. This force which causes free electrons to move in a particular direction in a conductor is known as electron moving force or electromotive force (e.m.f). wherever there is an electric charge, there exist electric field around it. In an electric field, potential at a point is defined as the amount of work done in bringing a unit +ve charge from infinity to that point. The units of potential is volt and symbol is V or EMF. In electric circuits, the potential (or voltage) is always specified between two points, not at a single point. The difference between the

ential of two points is called potential difference. e point having higher potential is termed as +ve rminal and the point having lower potential termed as -ve terminal. If a conductor is nected between +ve and -ve terminals, the ential difference between the two terminals ll force the flow of current through the ductor. The direction of flow of current will from +ve to -ve terminal.

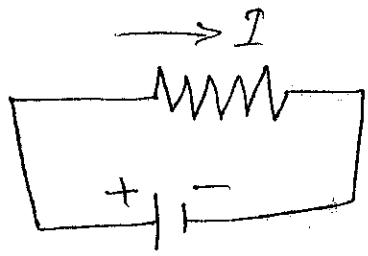


Fig. 2

HMS LAW

This law describes the relation between voltage and current (I) in a circuit. It states that rent (I) passing through a conductor is directly rtional to the potential difference (V) across two ends of a conductor provided the physical litions (i.e temperature, resistivity, dimensions etc) in the same.

Mathematically, ohm's law can be expressed as

$$V \propto I$$

$$\text{or } V = IR$$

where constant R is called resistance of the uctor and its unit is ohm. Resistance is the rty of the conductor which opposes the flow rrent. For any conductor, resistance R is given

$$R = \frac{\rho \cdot l}{a}$$

where ρ is called resistivity of the conductor and it depends upon the conductor material. 'l' is length of the conductor and 'a' is cross-sectional area of the conductor.

The reciprocal of resistance (R) is called the conductance (G) of the conductor.

$G = 1/R$ The unit of G is mho. Good conductors have higher values of conductance. The reciprocal of resistivity (ρ) is called conductivity (σ) of the conductor.

If a voltage V is applied across any resistance R then the voltage V is dropped across the resistance R and this drop in voltage will be equal to the applied voltage. This is also called drop in potential and is given by $V = IR$.

whenever there is drop in potential across any conductor then there will be loss of electric power across that conductor. suppose a voltage V is applied across a resistance R which is carrying a current I.

The loss of electrical power across resistance R is given by $P = V \times I$

$= (IR) \times I = I^2 R$

$= V \times \frac{V}{R} = \frac{V^2}{R}$

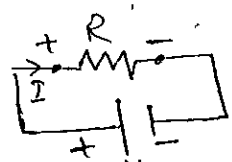


Fig-3

So loss in electrical power can be calculated as $P = I^2 R$ and also as $P = V^2/R$.

The terminal where current enters will be +ve terminal and the terminal where current leaves will be −ve terminal.

In any circuit element, if current I enters the +ve terminal of the element, then that element will consume electric power. If current I leaves the

ive terminal of the element, then power is delivered by the element.

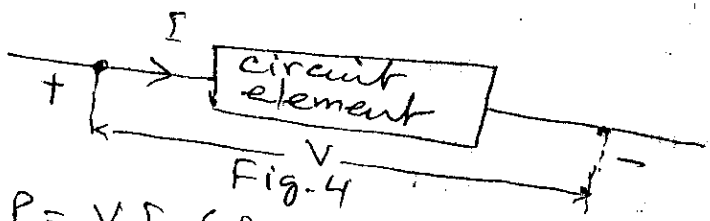


Fig. 4

$P = VI$ (Power absorbed by the element)

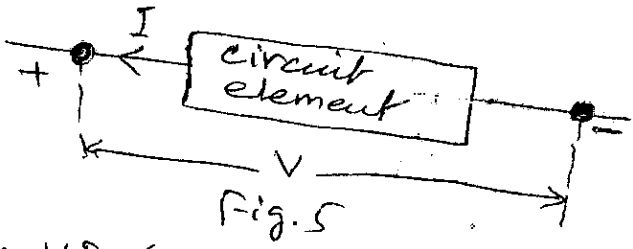
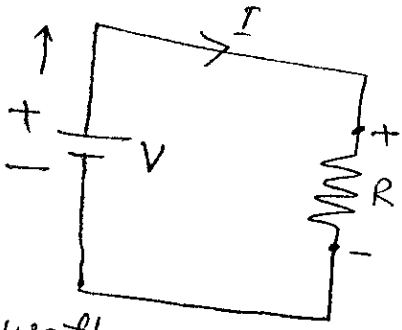


Fig. 5

$P = VI$ (Power delivered by the element).



0 watt delivered

10 watt consumed

Suppose $V = 5$ volt and $I = 2$ amp. then $P = VI = 2 \times 5 = 10$ wa

Fig. 6

CLASSIFICATION OF NETWORK ELEMENTS

Interconnection of some elements by a conductor is known as an electric circuit. The elements which supplies electric energy are called energy sources such as generator, battery etc. There are some elements which do not delivers electric energy and such elements consume or convert electric energy, such as Resistors, Inductors, capacitors etc.

Depending upon characteristics and properties, the circuit elements can be classified as follows.

(a) Active Elements :- The circuit elements which delivers or supply electric energy are called active elements. such as generators, batteries etc.

(b) Passive Elements :- The circuit elements which consume or absorb electric energy are called passive elements. such as resistors, inductors, capacitors etc.

(c) Linear Elements :- Linear elements are those elements in which response is directly proportional to the excitation. Examples of linear elements are pure resistors, Inductors and capacitors.

(d) Non-linear elements :- The elements in which ~~are not~~ response is not proportional to the excitation are called non-linear elements. such as diodes, semiconductors etc.

(e) Bilateral Elements :- The elements in which response to excitation do not depend on direction are called Bilateral elements. For example if a voltage V is applied across a conductor R, then

ulting current I will be V/R. If direction of the plied voltage is reversed then same current will flow through the resistor R. The resistors, ductors and capacitors are examples of bilateral ements.

Unilateral Elements:- Those elements in which sponse to excitation depends on direction of pplied excitation are called unilateral elements. r example when a diode is forward biased, conducts and when it is reversed biased it. oes not conduct.

, Lumped Elements:- The elements which are pure and separable are called lumped elements. For example ~~pure~~ pure resistors, inductors and capacitors.

) Distributed Elements:- the elements which are not pure and can not be separated are called distributed elements. For example transmission lines.

) Node:- Any point in the circuit where two or more elements are connected together is called a node. In the fig. shown, points a, b, c & d are node points.

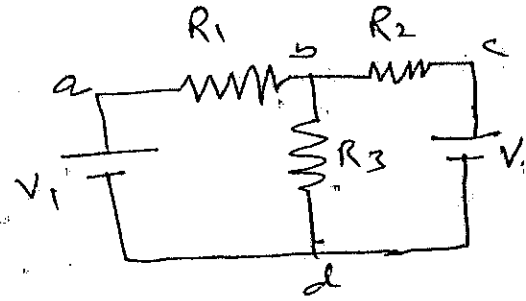


Fig.7

(j) <u>Branch</u>:- Any part of the circuit which lies between two nodes is called a branch. For example in the Fig.7, 'ab', 'bc', 'bc' etc. are branches.

(k) <u>Loop</u>:- Any closed path in the circuit in which one element or node is considered one time only is called loop. In Fig.7, there are only three loops, 'abda', 'dbcd' and 'abcda'.

(l) <u>Mesh</u>:- Any loop which contains no other loop is called Mesh. In Fig.7 loops 'abda' & 'dbcd' are called Mesh, but third loop 'abcda' is not a Mesh because it contain two loops i.e 'abda' & 'dbcd'.

ENENGY SOURCES:-

Depending upon the mode of current, ~~and~~ the electrical energy sources are classified as D.C source and A.C source. D.C (Direct current) source is that source in which magnitude of voltage or current do not change with time. whereas, in case of A.C electrical energy source, magnitude of current & voltage changes with time.

In this chapter only D.C sources will be discussed.

D.C electrical energy sources can be classified as ~~a~~ voltage source and ~~an~~ current source.

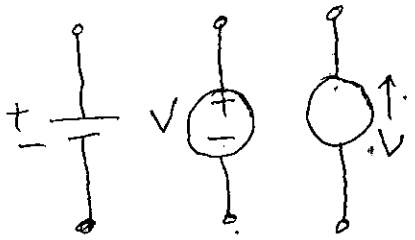
~~A.C~~ voltage source

- Ideal voltage source
- Practical voltage source
- Dependent voltage source
- Independent voltage source

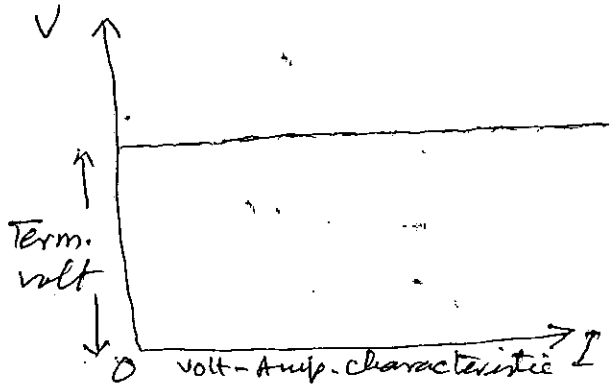
~~A.C~~ current source

- Ideal current source
- Practical current source
- Dependent current source
- Independent current source

<u>Ideal Voltage Source</u>:- Any voltage source (Battery or Generator) in which terminal voltage never changes with supplied current is called ideal voltage source.



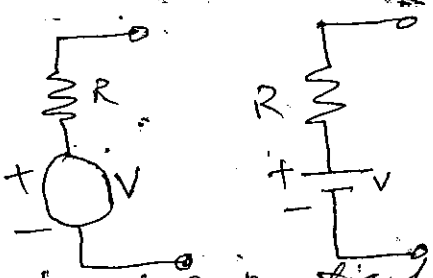
symbols of ideal voltage source.



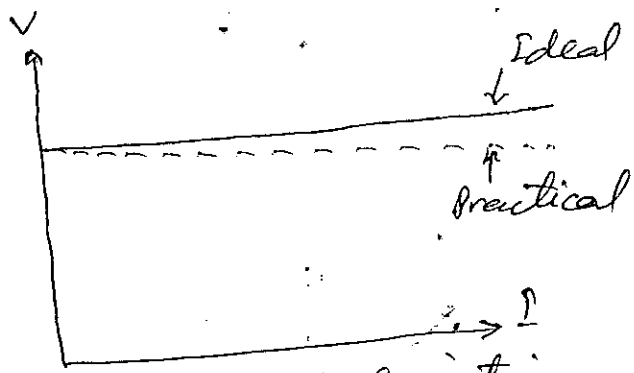
Volt-Amp. characteristic

In this world, nothing is ideal and hence an ideal voltage source is not possible. Therefore if current in the circuit increases, terminal voltage will ~~be~~ decrease. This is due to internal resistance of the voltage source.

<u>Practical Voltage Source</u>:- when current I in the circuit increases, there will be corresponding decrease in terminal voltage V due to internal resistance R. Such an voltage source is called practical voltage source.



symbol of practical voltage source



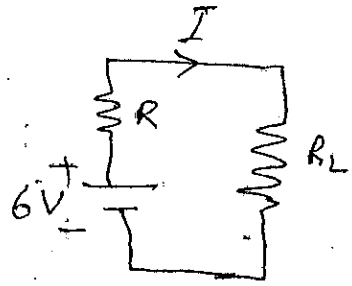
Volt-Amp. characteristic

In practical voltage source, as load current goes on increasing, terminal voltage goes on decreasing

Example:- A 6 volt battery has an internal resistance 0.00002 Ω. If load current is 100 Ampere. Calculate drop in voltage.

Soln:- V = 6 volt

R = 0.00002 Ω, I = 100 Amp.



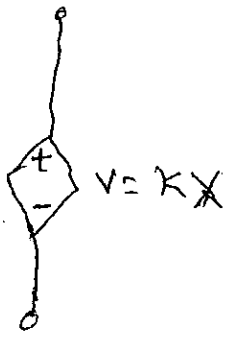
Internal voltage drop = 100 × 0.00002 = 0.002 V

Hence drop in voltage = 6 − 0.002 = 5.998 volt.

So if load current increases from 0 to 100 Amp., terminal voltage will drop only 0.002 volt. So it is almost an ideal voltage source.

Dependant voltage source:-

If the terminal voltage or current of the source depends on voltage of a certain element of the network, then the source is called dependent voltage source.



mbol of dependant
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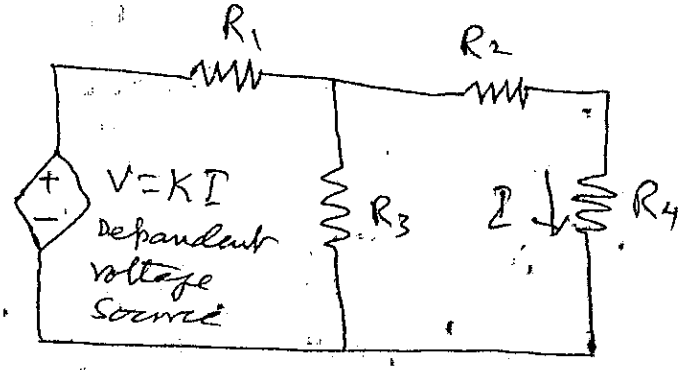
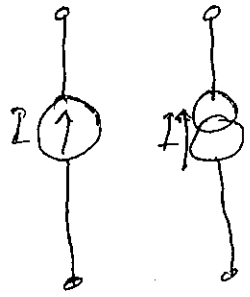


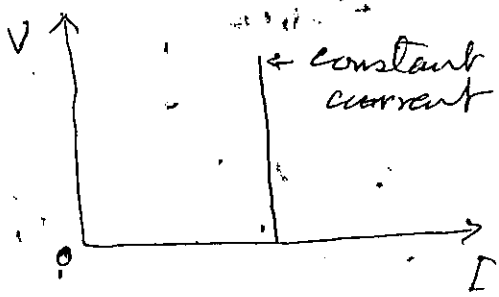
In the above fig., V depend on current in resistor R_4 i.e I.

<u>Independent voltage source</u>:- If the terminal voltage do not depend upon the voltage or current of any element in the circuit, then the source is called independent voltage source.

<u>Ideal current source</u>:- A current source which supplies constant current in the circuit irrespective of voltage across its terminals is called an ideal current source.

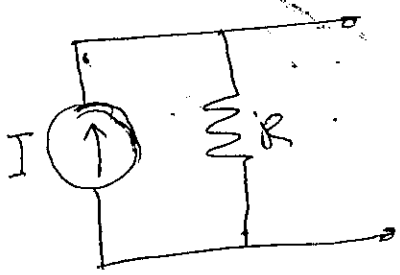


symbols of ideal current source

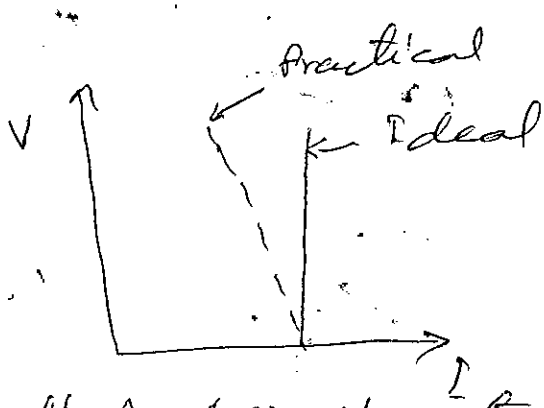


volt-Ampere characteristic of ideal current source.

<u>Practical current source</u>:- Ideal current source can not be possible. At every voltage level, constant current is not possible. Therefore in practice an ideal voltage source having internal resistance R in parallel exist.



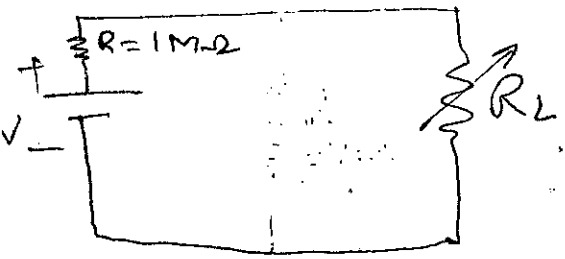
Practical current source



volt-Ampere characteristic of practical current source

If internal resistance of the current source is very high as compared to the load resistance then almost constant current can be maintained at all voltage levels.

<u>Example</u>:- A battery having terminal voltage 12 volt is having internal resistance of 1 MΩ. If load resistance varies from 20 KΩ to 200 KΩ, calculate variation in current.



Soln:- $V = 12$ volt

$R = 1\,M\Omega = 10^6\,\Omega$

$R_L = 20 - 200\,K\Omega$

at 20 KΩ, total resistance $= 1M\Omega + 0.02\,M\Omega$
$= 1.02\,M\Omega$

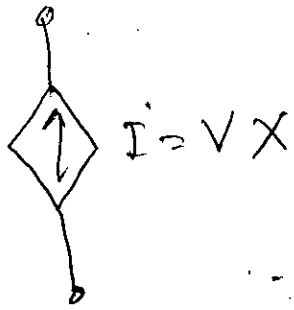
$$I = \frac{12}{1.02} = 11.8\,\mu A$$

at 200 KΩ, total resistance $= 1M\Omega + 0.2\,M\Omega$
$= 1.2\,M\Omega$

$$I = \frac{12}{1.2} = 10\,\mu A$$

Hence if load resistance increases 10 times, variation in voltage is only from 11.8 to 10 μA. This is nearly constant.

<u>Dependent Current Source</u>:- If the current of the current source depend upon the voltage or current of any circuit element, then such a current source is called dependent current source.



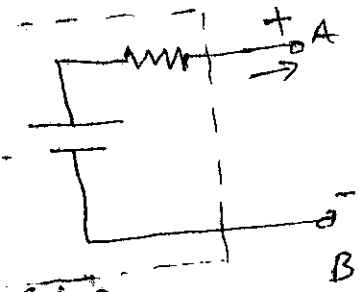
Symbol of dependent current source.

<u>Independent Current Source</u>:- If the current of the source does not depend upon the voltage or current of any element of the circuit, then such a source is called Independent Current Source.

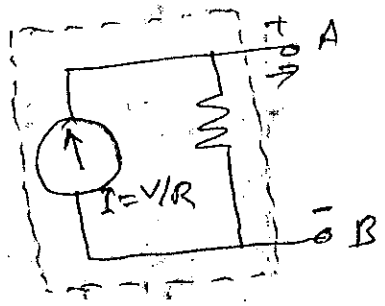
SOURCE CONVERSION

A voltage source can be converted to current source & a current source can be converted to a voltage source. But only practical sources can be converted.

A voltage source V having a resistance R in series can be converted to a current source $I = V/R$ having resistance R in parallel.



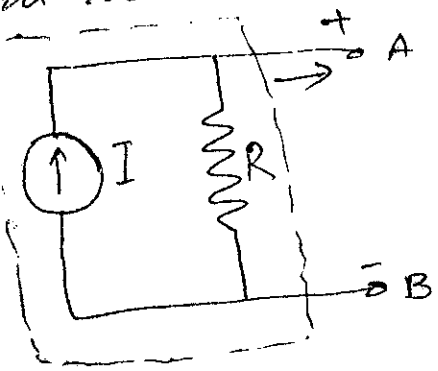
Given voltage source



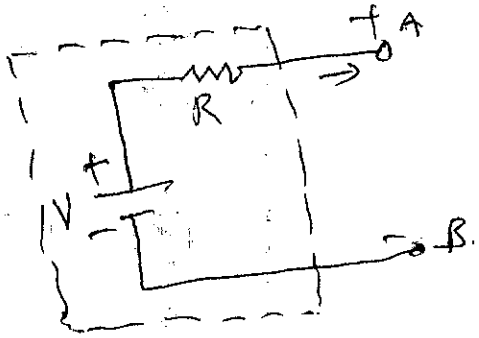
Equivalent current source

The direction of current should not be changed at the terminals, i.e. current should leave +ve terminal.

Similarly a practical current I source can be replaced having a resistance R in parallel to a voltage source $V = IR$, having resistance R in series.



Given current source



Equivalent voltage source.

Example :- using source transformation technique, determine the value of current through 15 Ω resistor.

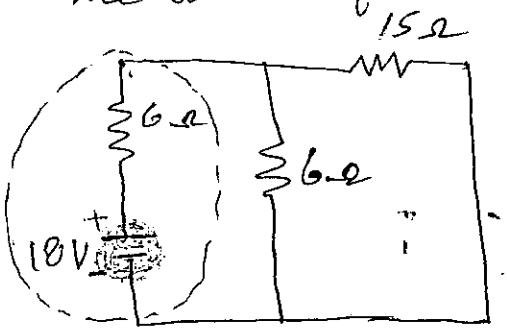


Fig. 1

$I = V/R = 18/6 = 3$ Amp.

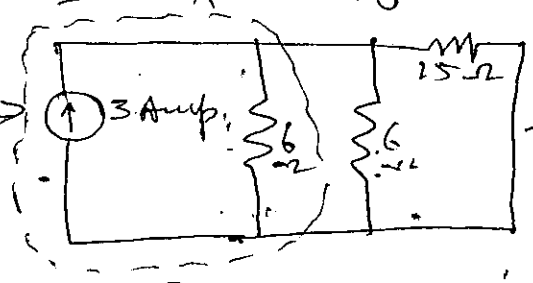


Fig. 2

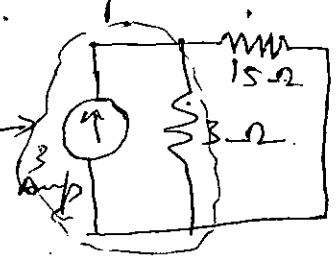
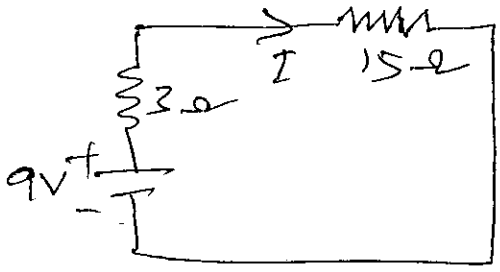


Fig. 3

current source of Fig. 3 can be converted to a voltage source $V = I \times R = 3 \times 3 = 9$ volt with resistance 3 Ω in series.

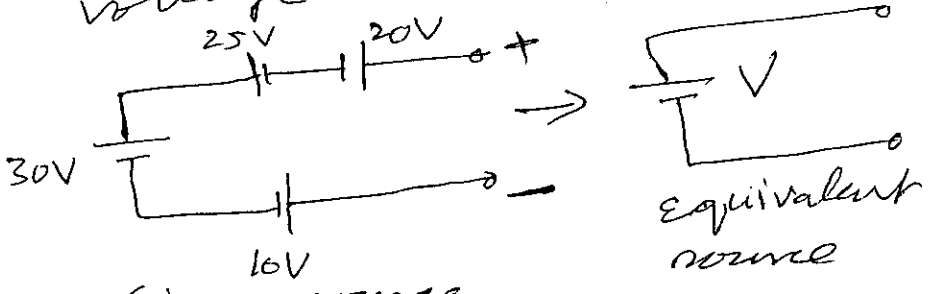


current flowing through 15 Ω resistance $I = \frac{V}{R} = \frac{9}{3+15} = \frac{9}{18} = 0.5$ Amp. Ans.

SOURCE COMBINATION

voltage sources can be connected in series and their equivalent source will be equal to sum of all sources.

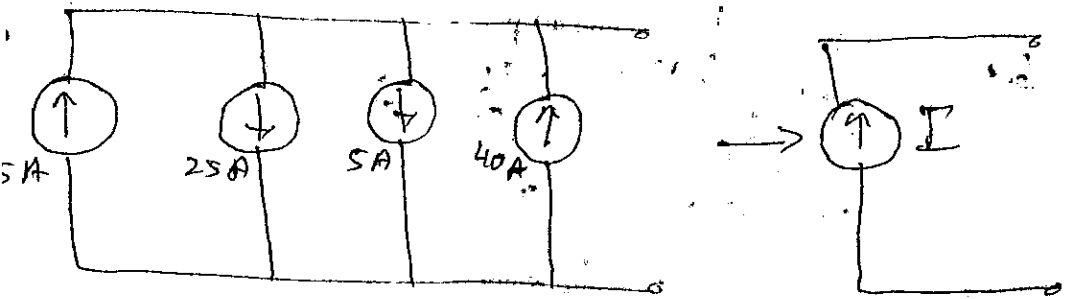
Example :- Four practical voltage sources having terminal voltage 30V, 25V, 20V + 10V are connected in series as shown below. calculate their equivalent voltage source



$V = 20 - 25 + 30 - 10$
$= 15$ volt Ans

current sources can be connected in parallel and their equivalent will be equal to sum of all current sources

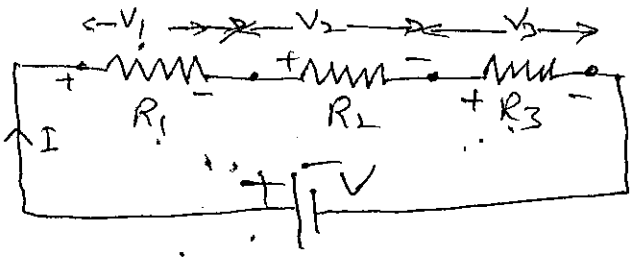
Example:-



$I = 15 - 25 - 5 + 40 = 25\ A.$ Ans

Combination of Resistors

An electrical circuit may have number of resistors. These resistors may be connected in different ways. If one end of a resistor is connected to one end of another resistor only, then such a combination of resistors is called <u>RESISTORS IN SERIES.</u>



If the resistors are connected in series as shown above, the same current will pass through each resistor. Voltage drop across each resistor will be different if each resistor has different resistance. The voltage drop will be directly proportional to the resistance. The applied voltage V will be equal to sum of all voltage drop across each resistor.

i.e $V = V_1 + V_2 + V_3$

$= IR_1 + IR_2 + IR_3$

$= I(R_1 + R_2 + R_3)$

or $\frac{V}{I} = R_1 + R_2 + R_3 = R_{eq}$ (Equivalent resistance).

Since same current I flows through each resistor, hence voltage drop against each resistor is directly proportional to its resistance

i.e $V_1 \propto R_1$, $V_2 \propto R_2$ and $V_3 \propto R_3$.
voltage drop across against any resistor R_i can be calculated as

$$V_i = \frac{R_i}{R_{eq}} \cdot V$$

POINTS TO REMEMBER

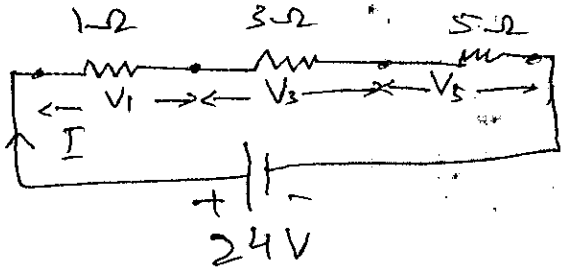
(i) current I through each resistor is same.

(ii) Voltage across each resistor is directly proportional to its resistance. The applied voltage is equal to sum of voltage drop against all resistors.

Example:- Three resistors having resistances 1Ω, 3Ω & 5Ω are connected in series. If applied voltage is 24 volt. calculate

(a) current through each resistor.

(b) voltage across each resistor.



$R_{eq} = 1+3+5 = 9\,\Omega$.

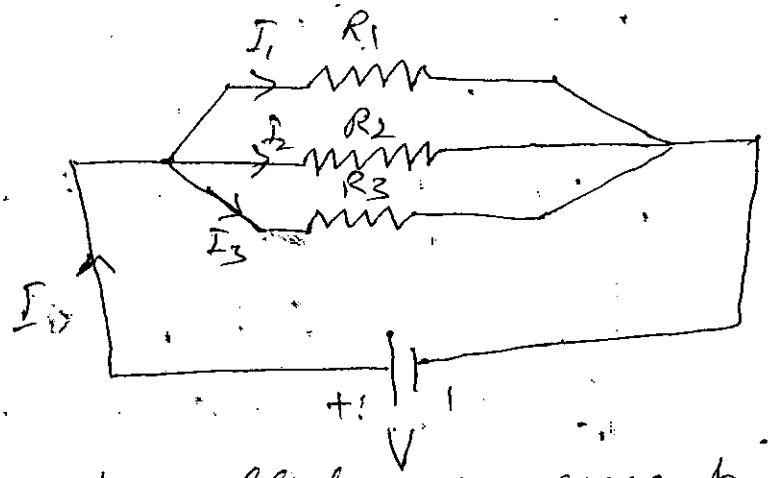
$$I = \frac{V}{R} = \frac{24}{9} = 2.66 \text{ Amp.}$$

$$V_1 = \frac{R_1}{R_{eq}} \cdot V = \frac{1}{9} \cdot 24 = 2.66 \text{ volt}$$

$$V_2 = \frac{R_2}{R_{eq}} \cdot V = \frac{3}{9} \cdot 24 = 8 \text{ volt}$$

$$V_3 = \frac{R_3}{R_{eq}} \cdot V = \frac{5}{9} \cdot 24 = 13.33 \text{ volt}$$

If one end of each resistor is connected to a common point and another end is also connected to a common point then such arrangement is called parallel connection.



In parallel arrangement of resistors, current is divided but voltage across each resistor is same i.e applied voltage V.

$$I = I_1 + I_2 + I_3$$

$$= \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3} = V\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right) = \frac{V}{R_{eq}}$$

Reciprocal of Resistance (R) is conductance (G).

Hence, $I = V \cdot G_1 + V \cdot G_2 + V \cdot G_3$

current through each resistor is directly proportional to conductance of each resistor.

i.e $I_i \propto G_i$

or $I_i \propto \frac{1}{R_i}$

$$1/R_{eq} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{R_2R_3 + R_1R_3 + R_1R_2}{R_1R_2R_3}$$

or $R_{eq} = \frac{R_1R_2R_3}{R_2R_3 + R_1R_3 + R_1R_2}$

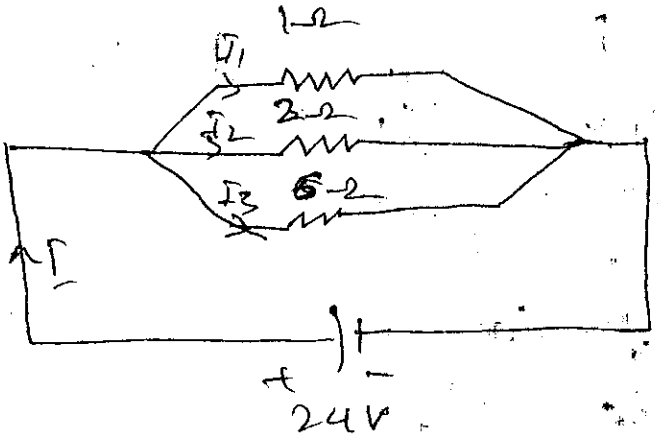
POINTS TO REMEMBER.

i) Voltage across each resistor remain same in parallel connection.

ii) Current through each resistor is directly proportional to its conductance or inversely proportional to its resistance.

iii) Total current is equal to sum of all individual current through all resistors.

Example :- If three unequal resistors of 1Ω, 2Ω & 6Ω are connected in parallel. Find current through each resistor and verify that sum of current through each resistor is equal to the applied current. Applied voltage is 24 volt.



$$I = I_1 + I_2 + I_3 = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$= V\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)$$

$$= 24\left(\frac{1}{1} + \frac{1}{2} + \frac{1}{6}\right)$$

$$= 24\,\frac{(6+3+1)}{6} = \frac{240}{6} = 40\,A$$

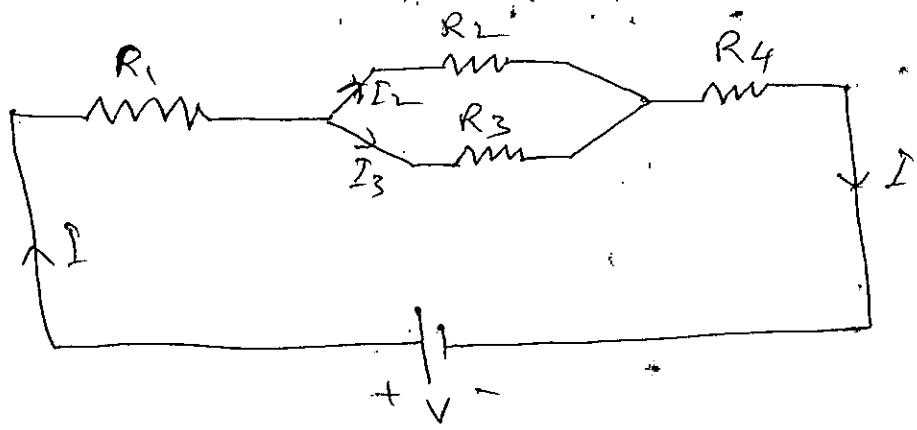
$$I_1 = \frac{V}{R_1} = \frac{24}{1} = 24\,A$$

$$I_2 = \frac{V}{R_2} = \frac{24}{2} = 12\,A$$

$$I_3 = \frac{V}{R_3} = \frac{24}{6} = 4\,A$$

SERIES PARELLEL CONNECTION OF RESISTORS

In an electrical circuit, resistors may be connected in combination of series and parallel as shown below.



$$R_{eq} = R_1 + \frac{1}{\frac{1}{R_2} + \frac{1}{R_3}} + R_4$$

$$R_{eq} = R_1 + \frac{R_2 R_3}{R_2 + R_3} + R_4$$

Example :- If $R_1 = 2\Omega$, $R_2 = 4\Omega$, $R_3 = 8\Omega$ and $R_4 = 1\Omega$. Find current I and current through R_2 and R_3. If applied voltage $V = 24$ volt.

$$R_{eq} = R_1 + \frac{R_2 R_3}{R_2 + R_3} + R_4 = 2 + \frac{4 \times 8}{4+8} + 1 = 2 + \frac{32}{12} + 1$$
$$= 5.66\ \Omega$$

$$I = \frac{V}{R_{eq}} = \frac{24}{5.66} = 4.24 \text{ Amp.}$$

$$I_2 = \frac{V_2}{R_{eq}}, \quad I_3 = \frac{V_3}{R_{eq}}, \quad V_2 = V_3 = V - (V_1 + V_4)$$

$$V_1 = I \times R_1 = 5.66 \times 2 = 11.32 \text{ volt}$$

$$V_4 = I \times R_4 = 5.66 \times 1 = 5.66 \text{ volt}$$

$$R_{eq} = \frac{4 \times 8}{4+8} = \frac{32}{12} = 2.66\ \Omega$$

$$V_2 = V_3 = V - (V_1 + V_4) = 24 - 16.98 = 7.02 \text{ volt}$$

$$\therefore I_2 = \frac{7.02}{4} = 1.755 \text{ Amp}, \quad I_3 = \frac{7.02}{8} = 0.877 \text{ Amp.}$$

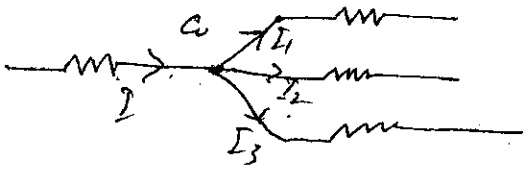
KIRCHHOFF's LAW's

To calculate voltages and currents in an electrical network, two laws namely kirchhoff current law (KCL) and Kirchhoff voltage law (KVL) are very important.

i) Kirchhoff's current law (KCL)

It states that at any node in the circuit, the sum of current entering is equal to the sum of current leaving. Or the sum of currents at any node is zero.

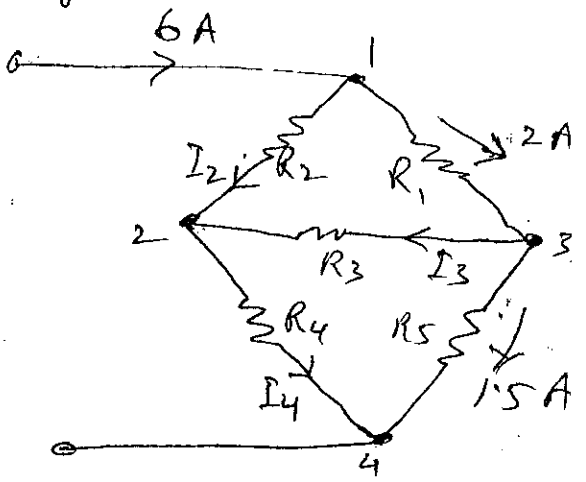
i.e $\sum_{i=1}^{N} I_i(t) = 0$



At node a shown in above fig., current I is entering and currents I_1, I_2 and I_3 are leaving.

i.e $I = I_1 + I_2 + I_3$ as per KCL.

Example :- calculate unknown currents in the diagram shown below.



Soln:- unknown currents are I_2, I_3 and I_4. There are four nodes in the circuit i.e 1, 2, 3 & 4.

Applying KCL at node 1.

$6 = 2 + I_2$, $\therefore I_2 = 6 - 2 = 4A$ p.

Applying KCL at node 3,

$2 = I_3 + 1.5$, $\therefore I_3 = 2 - 1.5 = 0.5A$ p

Applying KCL at node 2,

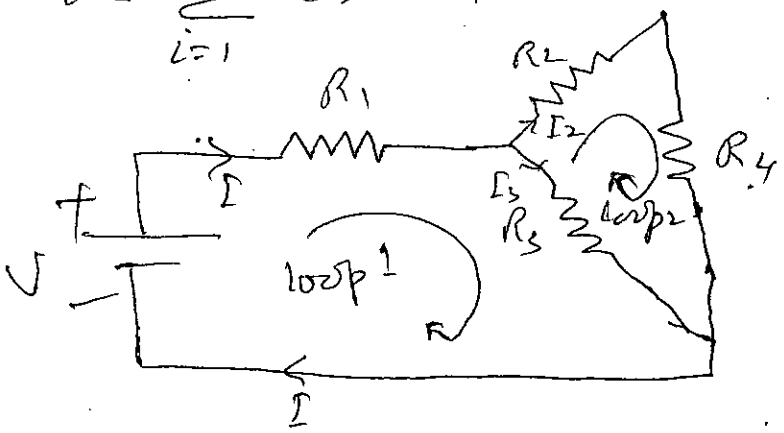
$2 + I_3 = I_4$ $\therefore I_4 = 4 + 0.5 = 4.5A$ p.

~~Applying KCL at node 4,~~

KIRCHHOFF'S VOLTAGE LAW (KVL)

According to this law, sum of voltages in a closed path is zero.

i.e $\sum_{i=1}^{N} V_i(t) = 0$



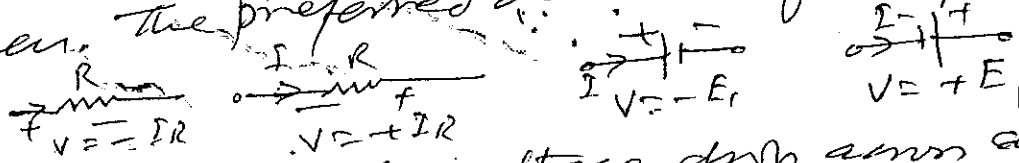
In loop 1 as shown above, if we apply KVL, then

$V - IR_1 - I_3R_3 = 0$

In loop 2, if we apply KVL, then

$-I_2R_2 - I_4R_4 + I_3R_3 = 0$

Sign convention :- If we move from +ve terminal of an element to −ve terminal, then negative voltage is taken and if we move from −ve terminal to +ve terminal of the element, then +ve voltage is taken. The preferred direction of loop is clockwise.



Example :- Calculate voltage drop across all resistors & verify KVL.

$I = \frac{V}{R_1+R_2+R_3} = \frac{12}{2+1+3} = 2$ Amp.

$V_1 = IR_1 = 2\times2 = 4$ volt

$V_2 = IR_2 = 2\times1 = 2$ volt

$V_3 = IR_3 = 2\times3 = 6$ volt

$-12 + 4 + 2 + 6 = 0$

STAR (Y) – DELTA (Δ) TRANSFORMATION

Electrical circuits are not only simple circuits. lectrical circuits can be very complicated also. o solve complicated circuits, star-delta or elta-star transformation is used. For example e circuit given below can not be solved usin imple series-parallel simplification method.

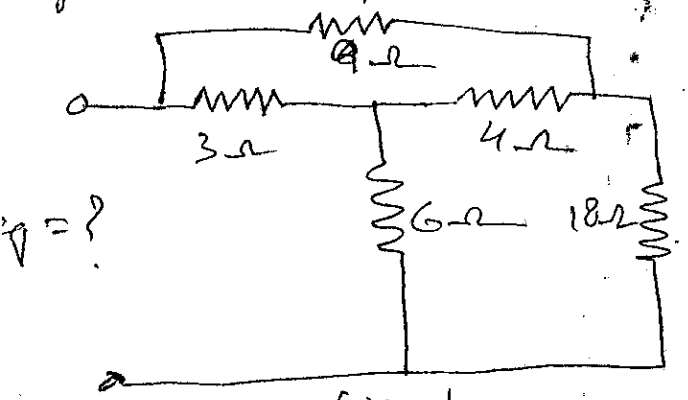


Fig-1

To solve such complicated circuits, star-delta transformation technique is used.

Delta to star transformation.

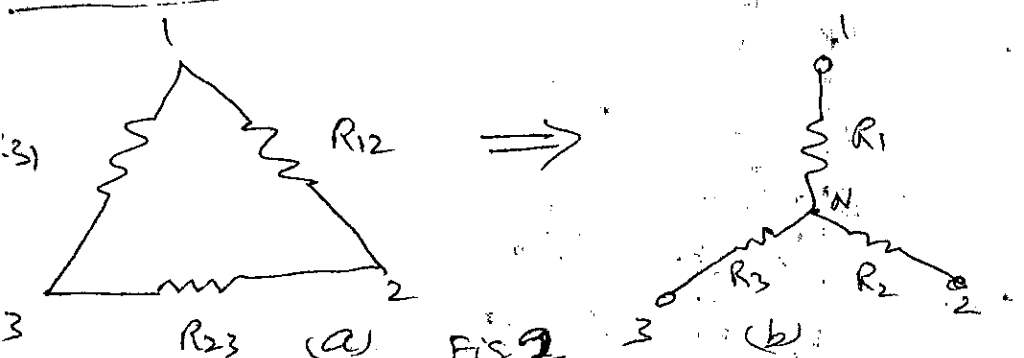


Fig 2

An star network has an common point called entral. Here given delta network is to be transformed to its equivalent star network.

In fig 2(b), resistance between terminal 1 & 2

$R_{12} = R_1 + R_2$ ——— (1)

because here third terminal is open.

whereas in fig 2(a), resistance between the terminals 1 & 2 will be R_{12} in parallel with

the series combination of $R_{31} + R_{23}$.

i.e $R_{12} = R_{12} \parallel (R_{23} + R_{31})$

$$= \frac{R_{12} \cdot (R_{23} + R_{31})}{R_{12} + R_{23} + R_{31}} \quad \text{—— (2)}$$

From eqns (1) & (2),

$$R_1 + R_2 = \frac{R_{12} \cdot (R_{23} + R_{31})}{R_{12} + R_{23} + R_{31}} \quad \text{—— (3)}$$

similarly between terminals 2 & 3, we have

$$R_2 + R_3 = \frac{R_{23} \cdot (R_{31} + R_{12})}{R_{12} + R_{23} + R_{31}} \quad \text{—— (4)}$$

and between terminals 3 & 1, we have

$$R_3 + R_1 = \frac{R_{31} (R_{12} + R_{23})}{R_{12} + R_{23} + R_{31}} \quad \text{—— (5)}$$

substracting eqn (4) from (3), (5) from (4) and (3) from (5) we have

$$R_1 - R_3 = \frac{R_{31} \cdot (R_{12} - R_{23})}{R_{12} + R_{23} + R_{31}} \quad \text{—— (6)}$$

$$R_2 - R_1 = \frac{R_{12} \cdot (R_{23} - R_{31})}{R_{12} + R_{23} + R_{31}} \quad \text{—— (7)}$$

$$R_3 - R_2 = \frac{R_{23} (R_{31} - R_{23})}{R_{12} + R_{23} + R_{31}} \quad \text{—— (8)}$$

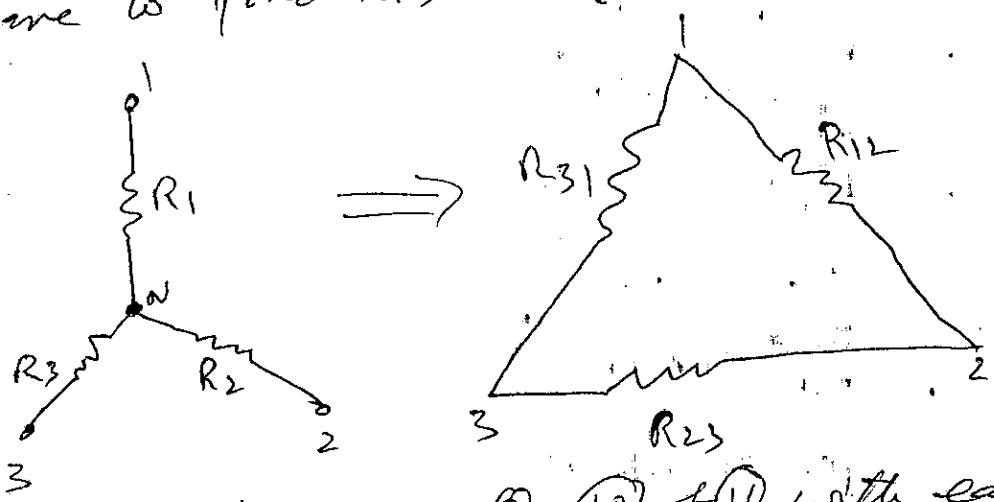
Adding eqns (5) & (6), we have

$$\cancel{2} R_1 = \frac{\cancel{2} R_{12} \cdot R_{31}}{R_{12} + R_{23} + R_{31}} \quad \text{or} \quad R_1 = \frac{R_{12} \cdot R_{31}}{R_{12} + R_{23} + R_{31}} \quad \text{—— (9)}$$

$$R_2 = \frac{R_{23} \cdot R_{12}}{R_{12} + R_{23} + R_{31}} \quad \text{—— (10)} \quad \& \quad R_3 = \frac{R_{23} \cdot R_{31}}{R_{12} + R_{23} + R_{31}} \quad \text{—— (11)}$$

) Star to Delta transformation

In this case R_1, R_2 & R_3 are given and we are to find R_{12}, R_{23} & R_{31}.



ultiplying eqns (9), (10) & (11) with each other & n addi, we get

$$R_1R_2 + R_2R_3 + R_3R_1 = \frac{R_{12}R_{23}R_{31}}{(R_{12}+R_{23}+R_{31})} \quad \text{——— (12)}$$

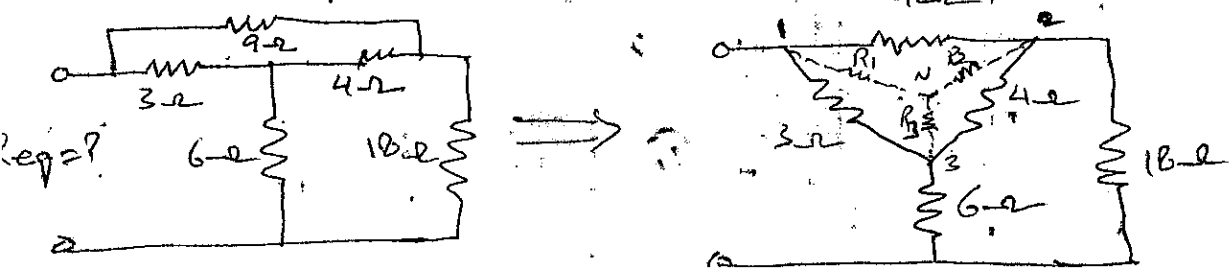
on dividing eqns (9), (10) & (11) by eqn (12) one by one, we get

$$R_{12} = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3} = R_1 + R_2 + \frac{R_1R_2}{R_3} \quad \text{——— (13)}$$

$$R_{23} = R_2 + R_3 + \frac{R_2R_3}{R_1} \quad \text{——— (14)}$$

$$R_{31} = R_1 + R_3 + \frac{R_1R_3}{R_2} \quad \text{——— (15)}$$

Exampled:- The circuit of fig 1 here can be solved using delta-star transformation technique.



using eqn (9), (10) & (11), we have

$$R_1 = \frac{R_{12} \cdot R_{31}}{R_{12}+R_{23}+R_{31}} \quad \text{—— (9)}, \qquad R_2 = \frac{R_{23} \cdot R_{12}}{R_{12}+R_{23}+R_{31}} \quad \text{—— (10)}$$

$$R_3 = \frac{R_{23} \cdot R_{31}}{R_{12}+R_{23}+R_{31}} \quad \text{—— (11)}$$

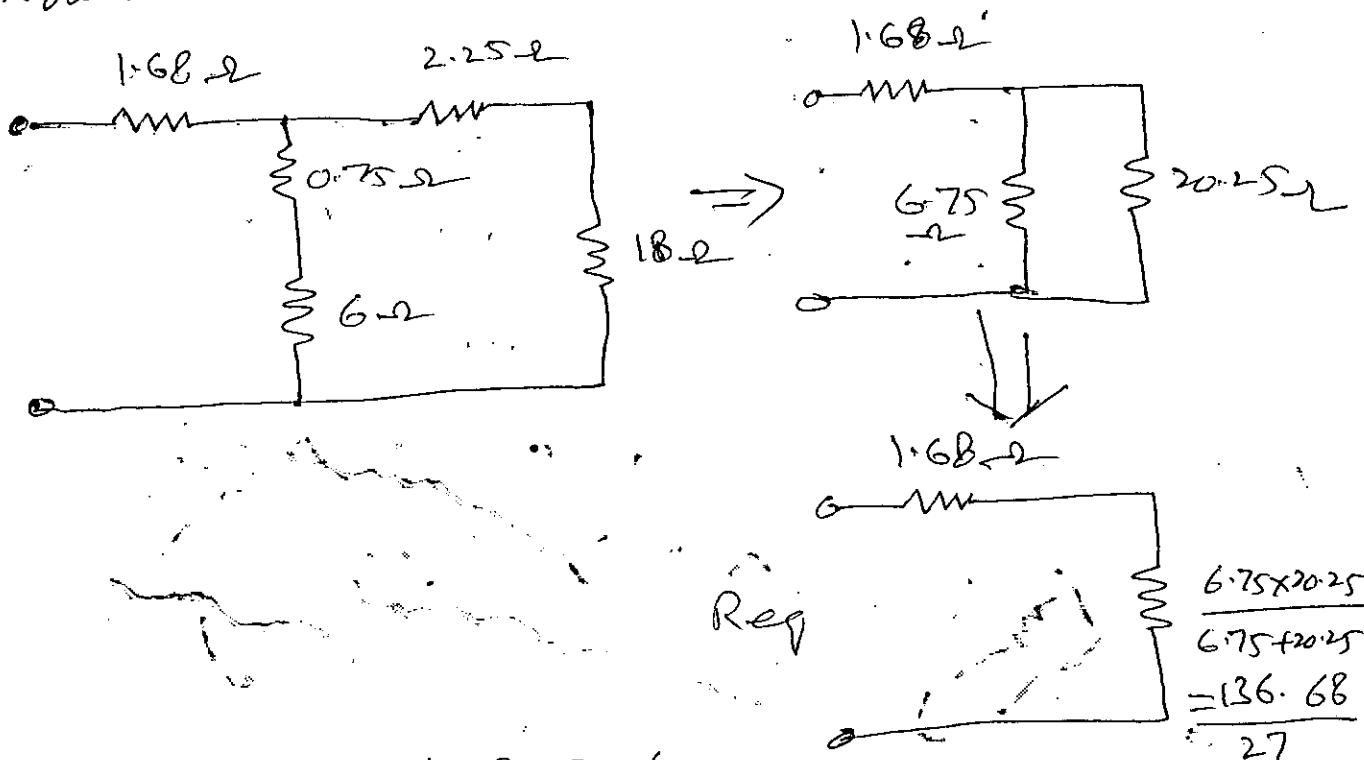
Here $R_{12} = 9\,\Omega$, $R_{23} = 4\,\Omega$ & $R_{31} = 3\,\Omega$

Therefore $R_1 = \frac{9 \times 3}{9+4+3} = \frac{27}{16} = 1.68\,\Omega$

$$R_2 = \frac{9 \times 4}{16} = \frac{36}{16} = 2.25\,\Omega$$

$$R_3 = \frac{4 \times 3}{16} = \frac{12}{16} = 0.75\,\Omega$$

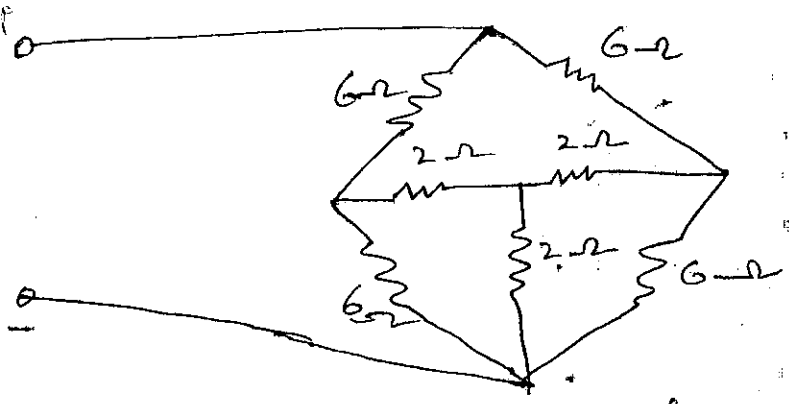
Now the circuit becomes,



Therefore $R_{eq} = 1.68 + 5.06$

$= 6.74\,\Omega$

Example-2 :- Solve the following to calculate equivalent resistance between terminals A + B.



First convert star connection of centre having 2, 2, 2 Ω resistors into delta.

Here $R_1 = 2\,\Omega$, $R_2 = 2\,\Omega$ + $R_3 = 2\,\Omega$

$$R_{12} = R_1 + R_2 + \frac{R_1 R_2}{R_3}, \quad R_{23} = R_2 + R_3 + \frac{R_2 R_3}{R_1}$$

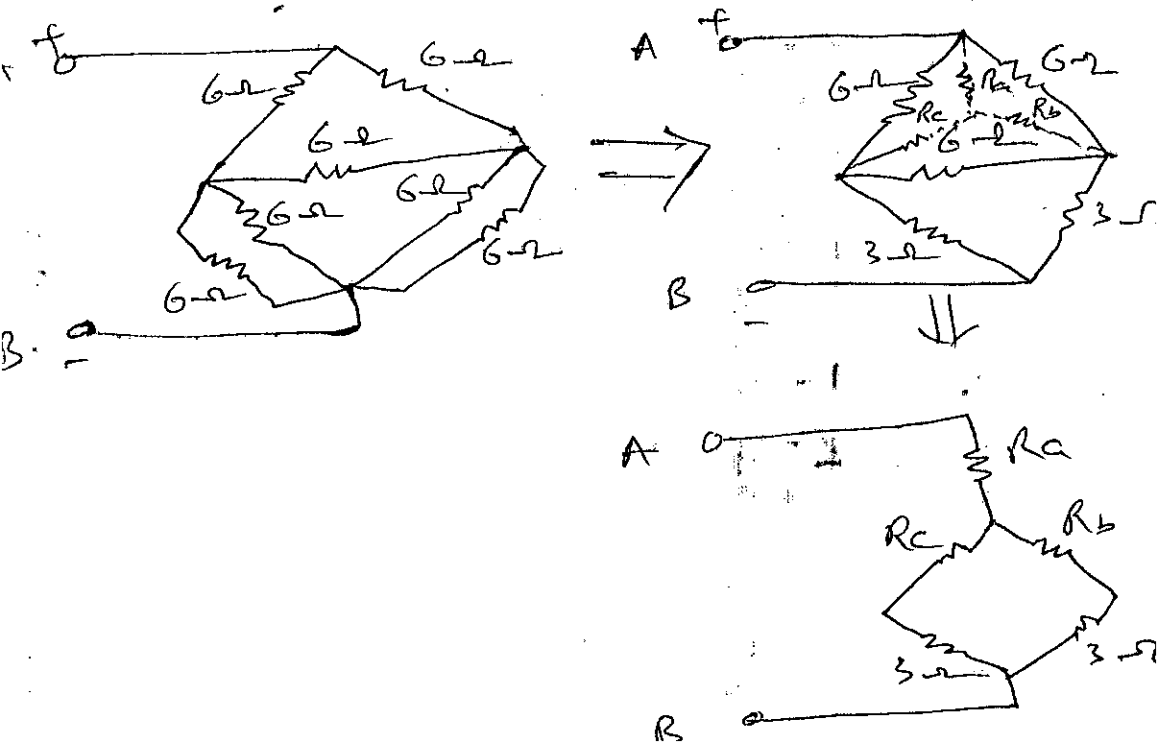
$$R_{31} = R_1 + R_3 + \frac{R_1 R_3}{R_2}$$

Therefore $R_{12} = 2 + 2 + \frac{2 \times 2}{2} = 4 + 2 = 6\,\Omega$

$R_{23} = 2 + 2 + \frac{2 \times 2}{2} = 4 + 2 = 6\,\Omega$

$R_{31} = 2 + 2 + \frac{2 \times 2}{2} = 4 + 2 = 6\,\Omega$

Therefore new network becomes.



UNIT – II (A.C Circuits)

(a) Sigle phase AC circuits.

Introduction

An alternating quantity (Alternating Current (AC) or Alternating voltage) is that quantity whose magnitude as well as direction changes with time.

An alternating current (AC) circuit is that circuit which is energized by alternating current or alternating voltage source.

Alternating current (AC) circuits are broadly divided into two categories

(a) Single phase AC circuits.
(b) Three phase AC circuits.

D.C circuits or D.C networks which we study in I unit are simple than AC circuits. Because in D.C networks there is only one element i.e resistor R other than energy source and magnitude of the parameters of D.C network do not change with time, whereas in A.C circuits, magnitude and direction of circuit parameters changes with time and also A.C circuits contain resistor R as well as inductor L and capacitor C in addition to energy sources.

No doubt, A.C circuits are complex, but A.C circuits have many advantages over D.C circuits and therefore all over the world A.C circuits ~~are~~ are

) in majority of applications.

The application or use of D.C system is very lied. D.C networks are being used in electronic its, charging of batteries, electric traction etc.

dvantages of A.C. system over D.C system.

The generation of A.C power is more efficient and economical.

The A.C voltage can be stepped up or stepped down more efficiently and this helps us in efficient transmission and distribution of electricity.

A.C motors are more cheap and efficient than D.C motors. This helps us to use electric power efficiently.

4) TV/radio signals are AC signals.

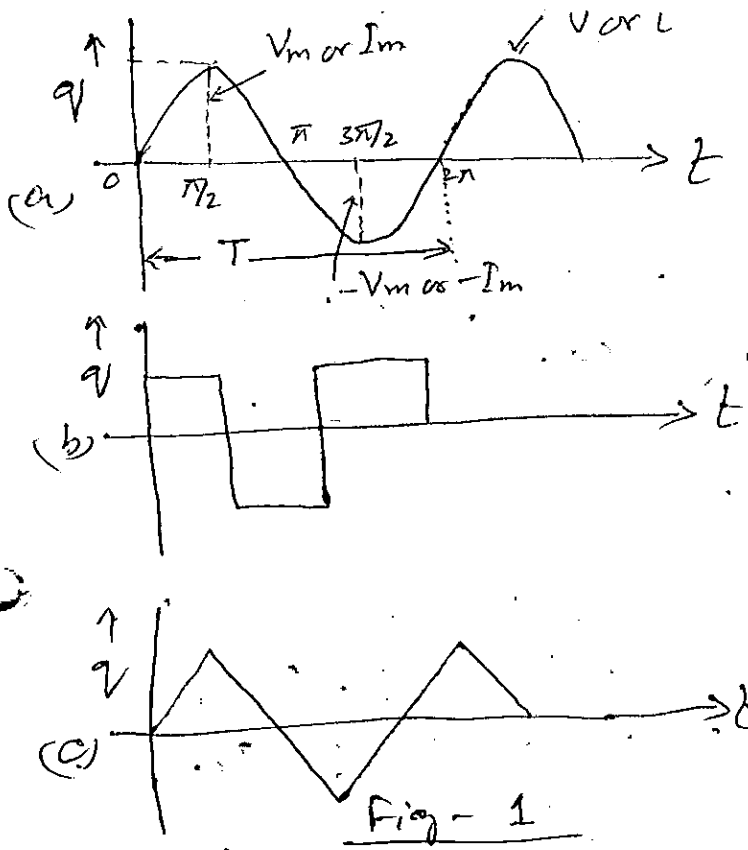
5) AC systems are universally adopted systems for generation, transmission and distribution of electricity.

Different terms related to Alternating quantity.

As stated earlier, an alternating quantity is that quantity whose magnitude changes with time. Or we can say that alternating quantity repeats its variation (wave shape) after a certain interval of time. Different terms related to A.C quantity are briefed as follows.

(1) Wave shape or wave form :- The figure which

we get by plotting instantaneous values with respect to time is called wave shape or wave form.



Sinusoidal wave alternating quantity.

square wave alternating quantity

Triangular wave alternating quantity.

Fig - 1

(2) <u>Instantaneous Value</u> :- The instantaneous value of any alternating quantity is defined as the value of the quantity at an instant of time. It can be determined from the curve or equation of the alternating quantity.

Example :- From curve or wave shape shown in Fig-1 (a) as shown above, value of alternating quantity at $t=0$ is 0 and at $t=\pi/2$ is V_m or I_m and at $t=\pi$ is 0.

If equation of an alternating current is $i = I_m \sin \omega t$ then from equation at $t=0$; $i=0$, at $t=\pi/2$ $(\omega=1)$, $\omega t=1$ $\therefore i = I_m$.

5) Maximum Value or Peak Value :- The maximum or peak value of an alternating quantity is the greatest magnitude of the quantity attained at any instant of time.

For example in Fig-1 (a), V_m or I_m is peak or maximum value of the curve.

6) Cycle :- A sine wave repeats itself after every 2π radians as shown in Fig-1 (a). Set of complete positive and negative values of the alternating quantity is called cycle.

7) Time period :- Time take by the alternating quantity to complete one cycle is called Time period and is denoted by T. shown in Fig-1 (a).

8) Frequency :- Frequency of an alternating quantity is defined as the number of cycles completed in one second. The unit of frequency is hertz (Hz). Same as cycles/second. In India system frequency is 50 Hz or 50 cycles/sec.

$$f = \frac{1}{T} \quad \text{or} \quad T = \frac{1}{f} = \frac{1}{50} = 0.02 \text{ sec}$$

9. Angular frequency (ω) :- A sine wave repeats itself after every 2π radians. If this quantity is divided by the time period; angular velocity or angular frequency is obtained.

i.e $\omega = \frac{2\pi}{T} = \frac{2\pi}{1/f} = 2\pi f$ radians/sec.

10) Phase :- Phase of any alternating quantity is defined as part of cycle or time period T passed from original position to any instant of time t.

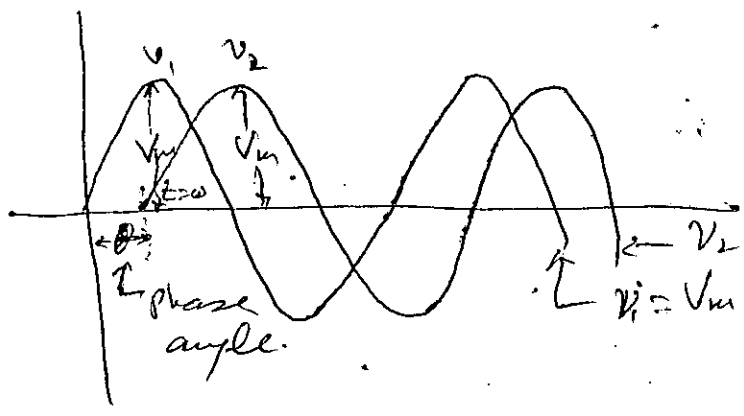
i.e phase at any instant of time t is given by t/T.

(9) <u>Phase Angle</u>:- Phase angle of an alternating quantity is defined as equivalent of phase expressed in radians or degrees. It is denoted by ϕ.

Therefore $\phi = \frac{2\pi t}{T} = \frac{2\pi t}{1/f} = 2\pi f t$.

The phase angle of any wave can also be defined as the angle of wave at $t=0$.

For example if an alternating voltage v is given by $v = V_m \sin(\omega t + \theta)$, then phase angle of this wave is $(\omega t + \theta)$. At $t=0$, phase angle becomes θ.



Here phase angle of wave v_1 is 0 and of v_2 is θ.

(10) <u>Phase difference</u>:- Phase difference between two waves is defined as the angular difference between two waves at $t=0$. As shown in fig. above, phase difference between V_1 & V_2 is θ.

Here v_1 starts at $t=0$

Phase difference between V_2 and V_1 is $\Delta\theta$.

$\Delta\theta = \theta - 0 = \theta$.

$\therefore$ If $\Delta\theta$ is +ve, v_2 is said to be leading v_1 by θ.

If $\Delta\theta$ is −ve, v_2 is " " " " lagging v_1 by θ.

xample :- Find the angle by which i_2 lags i_1 if

$i_1 = 120 \cos(100\pi t - 30^\circ)$ and

(i) $i_2 = -8\cos(100\pi t + 20^\circ)$

(ii) $i_2 = 5 \sin(100\pi t + 50^\circ)$

(iii) $i_2 = -6 \sin(100\pi t - 30^\circ)$

Soln:- i_1 can be written as

$$i_1 = 120 \sin(100\pi t - 30^\circ + 90^\circ)$$
$$= 120 \sin(100\pi t + 60^\circ)$$

$\therefore$ Phase angle of i_1 is 60°.

i.e $\theta_1 = 60^\circ$

(i) ~~Phase~~ $i_2 = -8\cos(100\pi t + 20^\circ)$
$$= 8 \sin(100\pi t + 20^\circ + 270^\circ)$$
$$= 8 \sin(100\pi t + 360^\circ - 70^\circ)$$
$$= 8 \sin(100\pi t - 70^\circ)$$

$\therefore$ Phase angle of i_2, $\theta_2 = -70^\circ$.

$\therefore$ i_2 lags i_1 by $= \theta_1 - \theta_2 = 60^\circ - (-70^\circ) = 130^\circ$

(ii) Phase angle of i_2 is $\theta_2 = 50^\circ$.

$\therefore$ i_2 lags i_1 by $= \theta_1 - \theta_2 = 60^\circ - 50^\circ = 10^\circ$

(iii) $i_2 = -6 \sin(100\pi t - 30^\circ)$
$$= -6 \cos(100\pi t - 30^\circ + 180^\circ)$$
$$= 6 \sin(100\pi t + 150^\circ)$$

Phase angle of i_2 is $\theta_2 = 150^\circ$.

$\therefore$ i_2 lags i_1 by $= \theta_1 - \theta_2$
$$= 60^\circ - 150^\circ$$
$$= -90^\circ$$

Since phase difference is -90°, therefore, we may also that i_2 leads i_1 by 90°.

Now suppose current I is passing through the same resistance R for the period t.

Therefore heat produced by D.C $I, H = \frac{I^2 \cdot R \cdot t}{J}$ —(2)

Equating eqns. (1) & (2), we have.

$$\frac{I^2 \cdot R \cdot t}{J} = \frac{Rt}{J}\left(\frac{i_1^2 + i_2^2 + i_3^2 + \cdots + i_n^2}{n}\right)$$

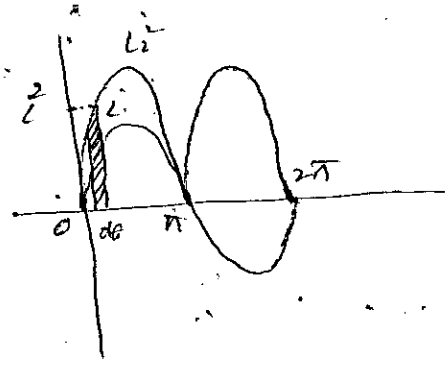
$$\therefore I^2 = \frac{(i_1^2 + i_2^2 + i_3^2 + \cdots + i_n^2)}{n}$$

$$\text{or } I = \sqrt{\frac{(i_1^2 + i_2^2 + i_3^2 + \cdots + i_n^2)}{n}}$$

or $I = \sqrt{\text{Mean (average) of the squares of all the instantaneous values of the current}}$

(2) <u>Analytical Method</u>:- In order to determine RMS value or effective value of an alternating current, we have to find squares of instantaneous values, then mean is taken and then square root is taken.

Let us consider a sine wave



$i = I_m \sin\theta$

$\therefore i^2 = I_m^2 \sin^2\theta$

Area of thin strip of thickness $d\theta$

is $= i^2 d\theta = dA$

$\therefore$ Area of half cycle $= \int_0^\pi dA = \int_0^\pi i^2 d\theta =$ Sum of squares of instantaneous values.

$\therefore$ Mean over half cycle $= \frac{\int_0^\pi i^2 d\theta}{\pi - 0} = \frac{\int_0^\pi i^2 d\theta}{\pi}$

$= \int_0^\pi I_m^2 \sin^2\theta \, d\theta$

$\therefore$ Root of the mean can be expressed as

$$I = \sqrt{\frac{\int_0^\pi I_m^2 \sin^2\theta \, d\theta}{\pi}} = I_m \sqrt{\frac{\int_0^\pi \sin^2\theta \, d\theta}{\pi}}$$

$$I = I_m\sqrt{\frac{1}{2\pi}\int_0^{\pi}(1-\cos 2\theta)\,d\theta}$$

$$= I_m\sqrt{\frac{1}{2\pi}\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi}}$$

$$= I_m\sqrt{\frac{1}{2\pi}\left[\pi - 0\right]} = I_m\sqrt{\frac{1}{2\pi}\cdot\pi} = \frac{I_m}{\sqrt{2}}$$

$$\therefore\ I = \frac{I_m}{\sqrt{2}}$$

$\therefore$ For a sine wave

$$\underline{\text{RMS Value} = \frac{1}{\sqrt{2}} \times \text{Maximum Value}}$$

In case of D.C, average value, RMS value, instantaneous value & peak value is same.

Form Factor

Ratio of RMS value to average value of an alternating quantity is defined as <u>Form Factor (K_f)</u>.

$$\text{i.e}\quad K_f = \frac{\text{RMS Value}}{\text{Average Value}}$$

$$= \frac{I_m/\sqrt{2}}{2I_m/\pi} = \frac{1/\sqrt{2}}{2/\pi} = \frac{1}{\sqrt{2}} \times \frac{\pi}{2} = 1.11$$

$\therefore$ $\underline{K_f = 1.11}$ for a sine wave.

Peak factor

Peak factor of a sine wave is defined as ratio of Maximum value to RMS value.

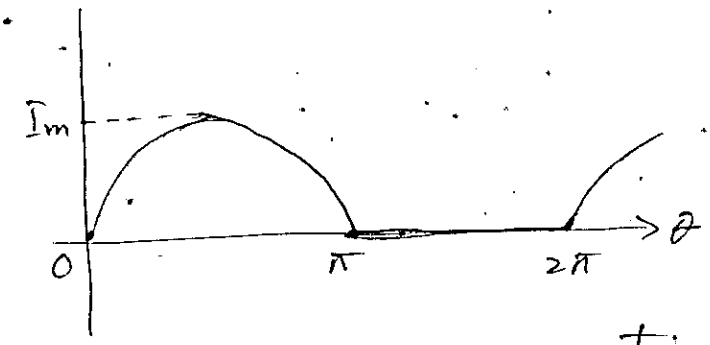
$$\text{i.e}\quad K_p = \frac{\text{Maximum Value}}{\text{RMS Value}} = \frac{I_m}{I} = \frac{I_m}{I_m/\sqrt{2}} = \sqrt{2}$$

$$\therefore\ \underline{K_p = 1.414}$$

for square wave $K_p = 1$ & for triangular wave $K_p = \sqrt{3}$

Example — Find RMS, average value & peak, form factor of a half rectified current wave (sinusoidal).

Soln:



The wave shown is unsymmetrical wave. Hence, RMS & average value will be calculated for complete cycle.

The equation of the wave is

$i = I_m \sin\theta$ for $0 \le \theta \le \pi$

$i = 0$ for $\pi \le \theta \le 2\pi$

$\therefore$ Average value $I_{av} = \frac{1}{2\pi}\int_0^{2\pi} i \, d\theta$

$$= \frac{1}{2\pi}\left[\int_0^{\pi} i\,d\theta + \int_{\pi}^{2\pi} i\,d\theta\right]$$

$$= \frac{1}{2\pi}\left[\int_0^{\pi} I_m \sin\theta\, d\theta + 0\right] = \frac{I_m}{2\pi}\left(-\cos\theta\right]_0^{\pi}$$

$$= \frac{-I_m}{2\pi}\left(+\cos\pi - \cos 0\right) = \frac{-I_m}{2\pi}(-1-1) = +\frac{I_m}{\pi}$$

$$= \frac{-I_m}{2\pi}(-2) = \frac{I_m}{\pi}$$

Similarly, RMS value

$$I^2 = \frac{1}{2\pi}\int_0^{2\pi} i^2 d\theta = \frac{1}{2\pi}\left[\int_0^{\pi} i^2 d\theta + \int_{\pi}^{2\pi} i^2 d\theta\right] = \frac{1}{2\pi}\left[\int_0^{\pi} I_m^2 \sin^2\theta\, d\theta + 0\right]$$

$$= \frac{I_m^2}{2\pi}\cdot\frac{1}{2}\int_0^{\pi}(1-\cos 2\theta)\, d\theta = \frac{I_m^2}{4\pi}\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi} = \frac{I_m^2}{4\pi}\left[\pi - 0\right] = \frac{I_m^2}{4}$$

$$\therefore I^2 = \frac{I_m^2}{4} \quad \therefore I = \sqrt{\frac{I_m^2}{4}} = \frac{I_m}{2}$$

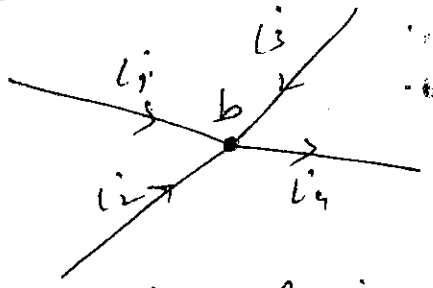
Form factor $K_f = \frac{I}{I_{av}} = \frac{I_m/2}{I_m/\pi} = \frac{\pi}{2} = 1.57$

Peak factor $K_p = \frac{I_m}{I} = \frac{I_m}{I_m/2} = 2$

Phasor representation of A.C quantities.

Any term which gives magnitude as well as phase an angle is called <u>Phasor</u> of an alternating quantity. Phasor representation of an A.C quantity greatly simplifies the calculations. Phasors are denoted by ~~bold~~ bold letters.

<u>Example:</u> Consider a node 'b' as shown in the fig. below.



The currents entering the node b are as given below.

$i_1 = 10 \sin \omega t$, $i_2 = 15 \cos \omega t$ & $i_3 = 20 \sin(\omega t - 45^\circ)$
$= 15 \sin(\omega t + 90^\circ)$

Calculate i_4.

Soln: Applying KCL at node b, we have

$$i_4 = i_1 + i_2 + i_3 = 10 \sin \omega t + 15 \cos \omega t + 20 \sin(\omega t - 45^\circ)$$

To calculate i_4, we need long trignometric calculation. The same problem can be solved very easily & quickly by phasor representation.

<u>Phasor representation of i_1, i_2 & i_3.</u>

i_1 can be represented by a phasor of magnitude 10 and phase angle zero along x-axis.

i.e. $\mathbf{I_{m1}} = I_{m1} \angle 0^\circ = 10 \angle 0^\circ$ ——— (1)

Similarly i_2 can be represented by a phasor of magnitude 15 and phase angle $+90^\circ$ along y-axis.

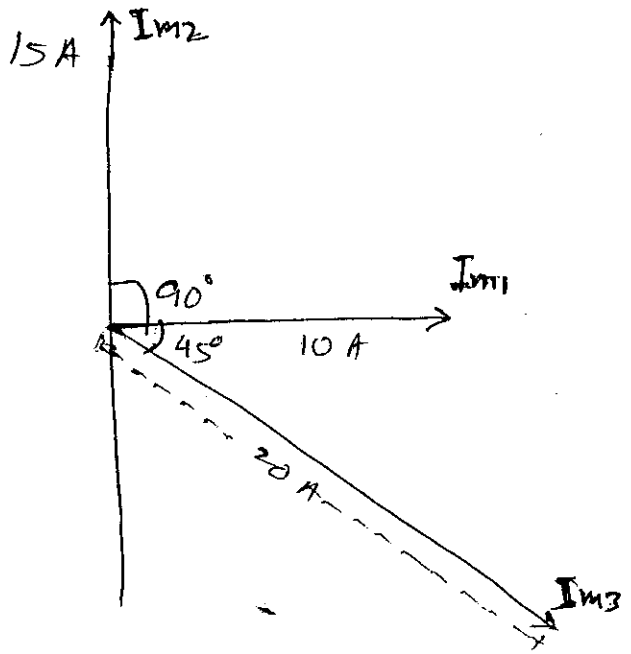
i.e. $\mathbf{I_{m2}} = I_{m2} \angle 90^\circ = 15 \angle 90^\circ$ ——— (2)

Similarly i_3, can be represented by a phasor of magnitude 20 and phase angle -45°

i.e. $\mathbf{I_{m3}} = I_{m3} \angle -45^\circ = 20 \angle -45^\circ$ ——— (3)

Equations (1), (2) & (3) gives phasors of maximum value of i_1, i_2 & i_3.

Phasors of i_1, i_2 & i_3 can be represented in terms of RMS values also.



Maximum value phasor representation.

Phasors of i_1, i_2 & i_3 can also be represented in terms of RMS values.

i_1 can be represented by a phasor of magnitude $\frac{10}{\sqrt{2}}$ and phase angle 0°.

i.e $\mathbf{I_1} = I_1 \angle 0° = \frac{10}{\sqrt{2}} \angle 0° = 7.07 \angle 0°$ —— (4)

where I_1 is RMS value of i_1 and $\mathbf{I_1}$ is phasor of i_1.

Similarly i_2 can be represented by a phasor of magnitude $\frac{15}{\sqrt{2}}$ and phase angle +90°

i.e $\mathbf{I_2} = I_2 \angle 90° = \frac{15}{\sqrt{2}} \angle 90° = 10.6 \angle 90°$ —— (5)

$\mathbf{I_2}$ is phasor of i_2 & I_2 is RMS value of i_2.

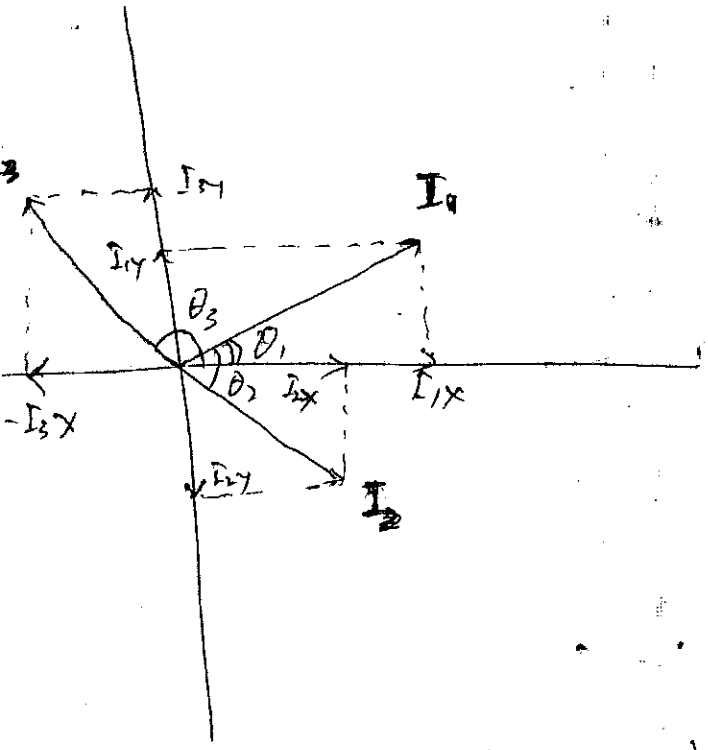
Similarly i_3 can be represented by a phasor of magnitude $\frac{20}{\sqrt{2}} = 14.14$ & phase angle $\angle -45°$.

i.e $\mathbf{I_3} = I_3 \angle -45° = \frac{20}{\sqrt{2}} \angle -45° = 14.14 \angle -45°$ —— (6)

Equations (4), (5) & (6) represents phasors of i_1, i_2 & i_3 in terms of RMS values of i_1, i_2 & i_3.

A phasor can be represented in the following forms.

1) Rectangular form.
2) Trignometric form.
3) Exponential form
4) Polar form.



Phasors I_1, I_2 & I_3 can be resolved along X & Y-axis.

The Y-axis is 90° ahead of X-axis.

i.e $Y = jX$, $j = 1\angle 90°$

$$I_1 = I_{1X} + jI_{1Y} \quad \text{——— (7)}$$

$$I_2 = I_{2X} - jI_{2Y} \quad \text{——— (8)}$$

$$I_3 = -I_{3X} + jI_{3Y} \quad \text{——— (9)}$$

Equations (7), (8) & (9) gives rectangular form representation of the phasors $\mathbf{I}_1$, $\mathbf{I}_2$ & $\mathbf{I}_3$ respectively.

From the fig., it is clear that

$I_{1x} = I_1 \cos\theta$ & $I_{1y} = I_1 \sin\theta$.

Putting these values in eqns (7), (8) & (9), we have

$$\mathbf{I}_1 = I_1 \cos\theta_1 + j I_1 \sin\theta_1 \quad \text{——— (10)}$$

$$\mathbf{I}_2 = I_2 \cos(-\theta_2) + j I_2 \sin(-\theta_2) \quad \text{——— (11)}$$

$$= I_2 \cos\theta_2 - j I_2 \sin\theta_2 \quad \text{——— (12)}$$

$$\mathbf{I}_3 = I_3 \cos\theta_3 + j I_3 \sin\theta_3 \quad \text{——— (13)}$$

Equations, (11), (12) & (13) gives trigonometric representation of phasors $\mathbf{I}_1$, $\mathbf{I}_2$ & $\mathbf{I}_3$.

from trignometry, we know that $e^{\pm j\theta} = \cos\theta \pm j\sin\theta$

Therefore eqns (11), (12) & (13) can be written as

$$\mathbf{I}_1 = I_1 e^{j\theta_1} \quad \text{——— (14)}$$

$$\mathbf{I}_2 = I_2 e^{-j\theta_2} \quad \text{——— (15)}$$

$$\mathbf{I}_3 = I_3 e^{j\theta_3} \quad \text{——— (16)}$$

Eqns (14), (15) & (16) gives exponential form representation of phasors.

Above eqns can also be written as

$$\mathbf{I}_1 = I_1 \angle\theta_1 \quad \text{——— (17)}$$

$$\mathbf{I}_2 = I_2 \angle -\theta_2 \quad \text{——— (18)}$$

$$\mathbf{I}_3 = I_3 \angle\theta_3 \quad \text{——— (19)}$$

Eqns (17), (18) & (19) gives polar form representation of phasors.

Applying KCL at node b, we have

$$I_4 = I_1 + I_2 + I_3$$

$$= 7.07 + j0 + 0 + j10.6 + 10 - j10$$

or $\boxed{I_4 = 17.07 + j0.6 \text{ A}}$

In polar form, we can write

$$I_4 = 17.08 \angle 2.01^\circ \text{ A}$$

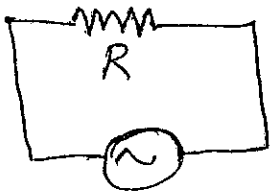
∴ Instantaneous value of i_4 is

$$i_4 = \sqrt{2} \times 17.08 \sin(\omega t + 2.01^\circ) \text{ A}$$

or $i_4 = 24.15 \sin(\omega t + 2.01^\circ)$ A.

RESPONSE OF CIRCUIT ELEMENTS TO AC VOLTAGE SOURCE

AC through Pure Resistance, R

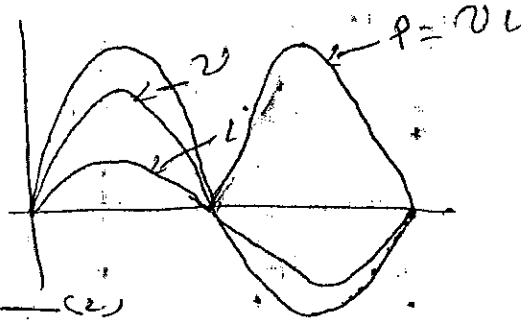


$$V = \frac{V_m}{\sqrt{2}} \angle 0^\circ$$

$$i = \frac{v}{R} = \frac{V_m \sin \omega t}{R}$$

$v = V_m \sin \omega t$ —(1)

or $i = I_m \sin \omega t$ —(2)



$I_m = \frac{V_m}{R}$ = Maximum value of current.

Phasor of voltage $V = \frac{V_m}{\sqrt{2}} \angle 0^\circ$

Phasor of current $I = \frac{I_m}{\sqrt{2}} \angle 0^\circ$

From equations (1) & (2), it is clear that in a pure resistance, voltage & current are in the same phase.

Instantaneous power consumed by the resistance is given by

$$p = v \cdot i = V_m \sin \omega t \cdot I_m \sin \omega t = V_m I_m \sin^2 \omega t$$

Power consumed in one cycle is (Average)

$$P = \frac{1}{2\pi} \int_0^{2\pi} V_m I_m \sin^2 \omega t \cdot d(\omega t) = \frac{V_m I_m}{2\pi} \int_0^{2\pi} \frac{(1 - \cos 2\omega t)}{2} d(\omega t)$$

Let $\omega t = \theta$

$$\therefore P = \frac{V_m I_m}{4\pi}\int_0^{2\pi}(1-\cos 2\theta)\,d\theta = \frac{V_m I_m}{4\pi}\left(\theta - \sin\frac{2\theta}{2}\right)_0^{2\pi}$$

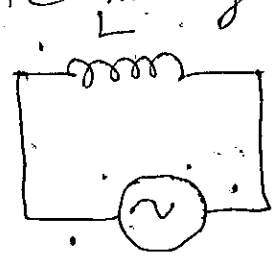
$$= \frac{V_m I_m}{4\pi}\times 2\pi = \frac{V_m I_m}{2} = \frac{V_m}{\sqrt{2}}\times\frac{I_m}{\sqrt{2}}$$

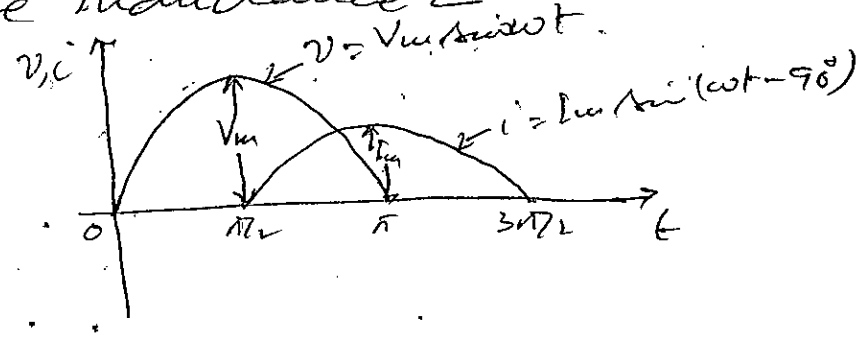
or $P = VI$ watts

where V & I are RMS values of applied voltage & current.

It is to be noted that instantaneous power is never negative and it is zero at $\theta = 0, \pi, 2\pi$ & maximum at $\theta = 90°$ & $270°$.

AC through pure Inductance L





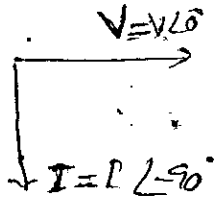
$$V = \frac{V_m}{\sqrt{2}}$$

When a voltage is applied across an inductor having inductance L henry, current starts flowing. The relationship between voltage & current is given by the equation.

$$v = L\frac{di}{dt} \quad \text{or} \quad di = \frac{v}{L}dt \quad \text{Integrating both sides}$$

we have $i = \int \frac{v}{L}dt = \int \frac{V_m \sin\omega t\,dt}{L} = \frac{V_m}{L}\int \sin\omega t\,dt$

$$\therefore i = \frac{V_m}{\omega L}(-\cos\omega t) = \frac{V_m}{\omega L}\sin(\omega t - 90°)$$



where $I_m = \frac{V_m}{\omega L}$ = Maximum value of current.

or $\omega L = \frac{V_m}{I_m}$

The term ωL is called inductive reactance and its unit is ohm. i.e $X_L = \omega L\ \Omega$.

$$\therefore i = \frac{V_m}{\omega L}\sin(\omega t - 90°) = I_m \sin(\omega t - 90°)$$

m above equ, it is clear that when a voltage is plied across a pure inductor, current lags voltage 90°.

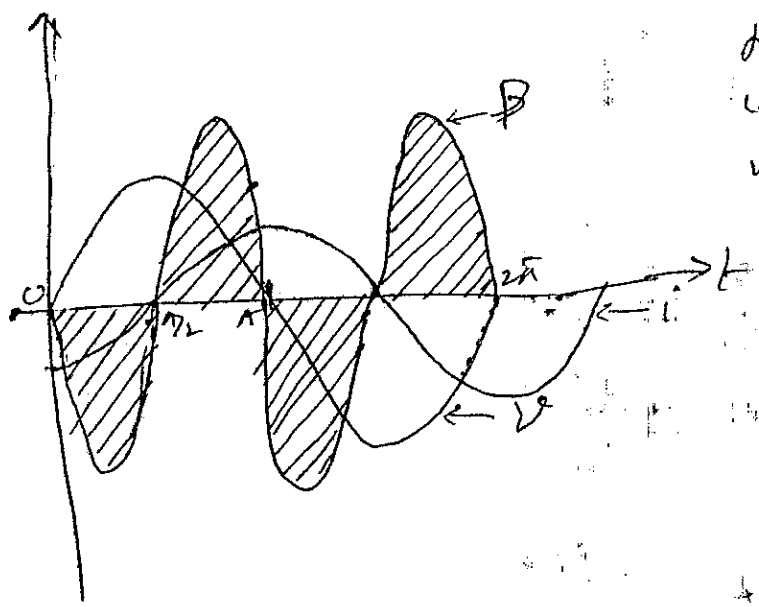
The instantaneous power consumed by the inductor given by

$$p = vi = V_m \sin\omega t \cdot I_m \sin(\omega t - 90°)$$
$$= V_m I_m \sin\omega t(-\cos\omega t)$$
$$= -V_m I_m \sin\omega t \cos\omega t$$
$$= -\frac{V_m I_m}{2} \sin 2\omega t$$

Therefore average power over a cycle is

$$P = \frac{1}{2\pi}\int_0^{2\pi} -\frac{V_m I_m}{2} \sin 2\omega t \cdot d(\omega t) = 0$$

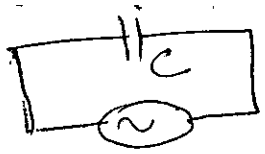
Therefore average power consumed by an inductor is zero over a cycle. However the frequency of a power wave is double the frequency of supplied wave. whenever supplied voltage and resulting current have same polarity, the power is positive and when polarity of supplied voltage and resulting current is opposite,



the power becomes negative. Positive power means, storage of electric energy in the form of magnetic flux. Negative power means release of electrical energy due to collapse of magnetic flux. Thus energy consumed is zero.

AC through a pure capacitor.

If a voltage wave $v = V_m \sin \omega t$ is applied across a pure capacitor of C farad, then resulting current starts flowing. The relation between applied voltage and resulting current is given by



$$i = C\frac{dv}{dt} = C\frac{d}{dt}(V_m \sin \omega t).$$

or $i = C V_m \omega \cos \omega t$

$$i = \frac{V_m}{\frac{1}{\omega C}} \sin(\omega t + 90^\circ)$$

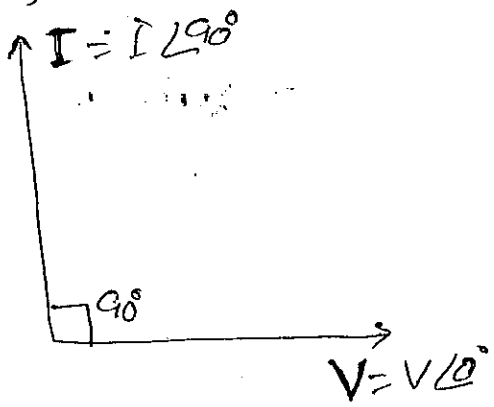
or $i = I_m \sin(\omega t + 90^\circ)$

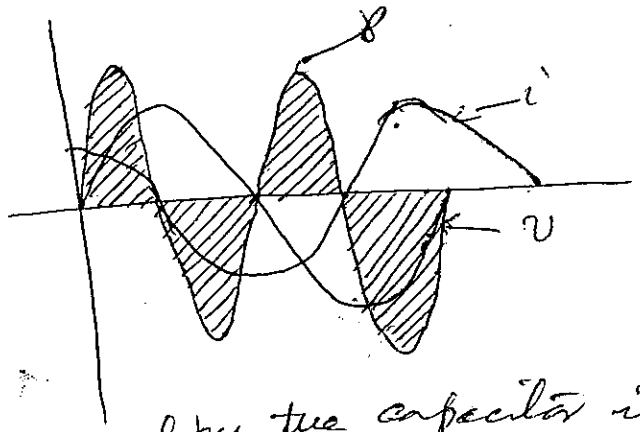
where $I_m = \frac{V_m}{\frac{1}{\omega C}}$ = Maximum value of current

$\frac{1}{\omega C} = \frac{V_m}{I_m}$ = Resistance and $\frac{1}{\omega C}$ is called capacitive reactance denoted by X_C

or $X_C = \frac{1}{\omega C}\ \Omega$.

From the eqn. of current wave $i = I_m \sin(\omega t + 90^\circ)$, it is clear that current leads voltage by 90°.





The instantaneous power consumed by the capacitor is given by $p = v \cdot i = V_m \sin \omega t \cdot I_m \sin(\omega t + 90^\circ)$

$$= V_m I_m \sin \omega t \cos \omega t = \frac{V_m I_m}{2} \sin 2\omega t$$

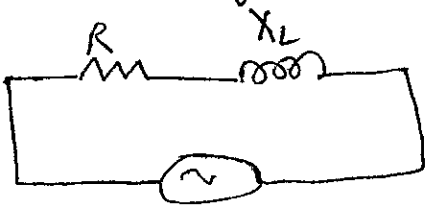
Average power over a cycle is given by

$$P = \frac{1}{2\pi}\int_0^{2\pi} \frac{V_m I_m}{2} \sin 2\omega t \, d(\omega t) = \frac{V_m I_m}{4\pi}\int_0^{2\pi} \sin \omega t \, d(\omega t) = 0$$

Hence $P = 0$, which means power consumed by a capacitor over a cycle is zero. However, frequency of power wave

is double of the supply frequency. Positive power means storage of electric energy in the form of electrostatic energy of the capacitor (charging). Negative power means, release of ~~electric energy~~ electrostatic energy (discharging) of capacitor.

AC Through R-L Series Circuits.

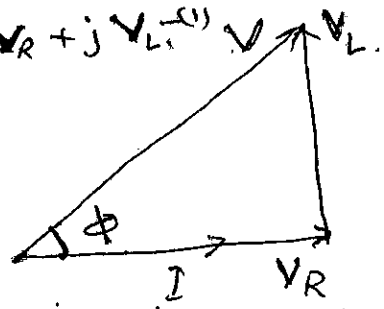


$v = V_m \sin(\omega t)$

or $\mathbf{V} = V\angle 0^\circ$

If a voltage v whose phasor value is $\mathbf{V}$ is applied across a series combination of resistor & inductor then a current i starts flowing. $i = I_m \sin(\omega t - \phi)$ or $\mathbf{I} = I\angle -\phi$. When current will flow, there will be voltage drop across the resistor having resistance R and inductor having inductive reactance X_L. Voltage drop across resistance $\mathbf{V}_R$ will be in phase of the current $\mathbf{I}$ and voltage drop across inductor $\mathbf{V}_L$ will lead current by 90°

i.e $\mathbf{V} = \mathbf{V}_R + j\mathbf{V}_L$ —(1)



Equation (1) is called voltage equation and diagram is called voltage triangle.

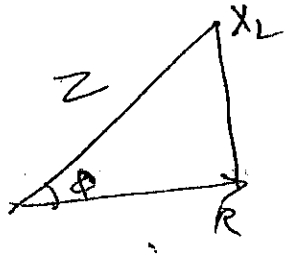
From the voltage diagram, $\tan\phi = \frac{V_L}{V_R}$ or $\phi = \tan^{-1}\left(\frac{V_L}{V_R}\right)$.

Equation (1) can also be written as

$\mathbf{V} = \mathbf{I}\cdot R + j\,\mathbf{I}\cdot X_L$ $\qquad V = \sqrt{V_R^2 + V_L^2}$

or $\frac{\mathbf{V}}{\mathbf{I}} = R + jX_L$

or $Z = R + jX_L$ —(2) where $Z = \frac{\mathbf{V}}{\mathbf{I}}$ = Impedance of the circuit.



Equation (2) is called impedance eqn & diagram is called impedance triangle.

$\tan\phi = \frac{X_L}{R}$ or $\phi = \tan^{-1}\frac{X_L}{R}$

$Z = \sqrt{R^2 + X_L^2}$

Power and Power triangle

$v = V_m \sin \omega t$, $i = I_m \sin(\omega t - \phi)$

$\therefore$ Instantaneous power $p = v \cdot i$

or $p = V_m \sin \omega t \cdot I_m \sin(\omega t - \phi)$.

$= V_m I_m \sin \omega t \cdot \sin(\omega t - \phi)$

$= \frac{V_m I_m}{2}[2 \sin \omega t \sin(\omega t - \phi)]$

$= \frac{V_m I_m}{2}[\cos\phi - \cos(2\omega t - \phi)]$

$= \frac{V_m I_m}{2} \cos\phi - \frac{V_m I_m}{2} \cos(2\omega t - \phi)$

$\therefore$ Average power over one cycle

$$P = \frac{1}{2\pi}\int_0^{2\pi} v \cdot i = \frac{1}{2\pi}\frac{V_m I_m}{2}\int_0^{2\pi} \cos\phi \cdot d(\omega t) - \frac{1}{2\pi}\frac{V_m I_m}{2}\int_0^{2\pi} \cos(\omega t - \phi) \cdot d(\omega t)$$

$$= \frac{1}{2\pi}\frac{V_m I_m}{2} \cdot 2\pi \cdot \cos\phi - 0 = \frac{V_m I_m}{2}\cos\phi = \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}} \cdot \cos\phi$$

or $\boxed{P = V I \cos\phi}$ —(3) where V & I are RMS values of applied voltage & resulting current.

ϕ is angle between applied voltage and resulting current and is called power factor angle. $\cos\phi$ is called power factor.

P is active power because this power does the useful work.

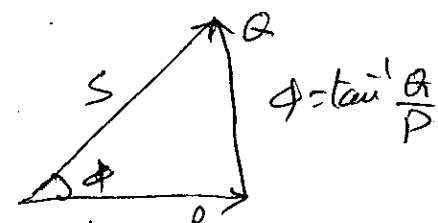
From eqn (3), it is clear that active power depends upon power factor $\cos\phi$.

Equation (1) is $\mathbf{V} = \mathbf{V}_R + j\mathbf{V}_L$

$= V\cos\phi + jV\sin\phi$

or $\mathbf{IV} = \mathbf{I}V\cos\phi + j\mathbf{I}V\sin\phi$

or $S = P + jQ$, $S = \sqrt{P^2 + Q^2}$



Power triangle.

where S is called apparent power. Its units is VA or KVA

P is called active power. Its unit is watt or kilowatt (kW)

Q is called reactive power. Its unit is VAR or KVAR.

Example:- A coil having a resistance of 3.5 Ω and inductance of 0.07 H is connected across 220V, 50 Hz supply. Find current drawn by the supply & power factor of the circuit.

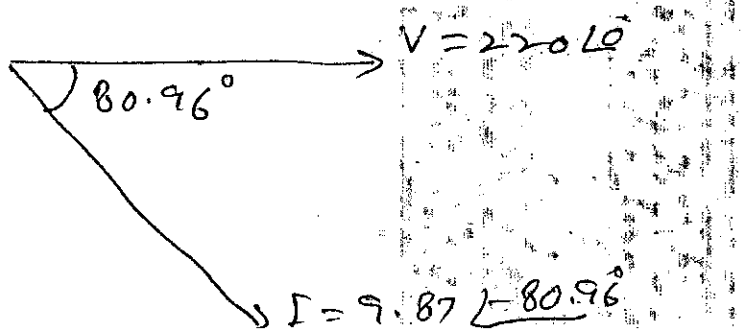
Soln:- $R = 3.5\,\Omega$, $L = 0.07\,H$, $V = 220\angle 0^\circ$

$$X_L = \omega L = 2\pi f L = 2 \times 3.14 \times 50 \times 0.07$$
$$= 22\,\Omega.$$

$$Z = R + jX_L = 3.5 + j22 = 22.276\angle 80.96^\circ$$

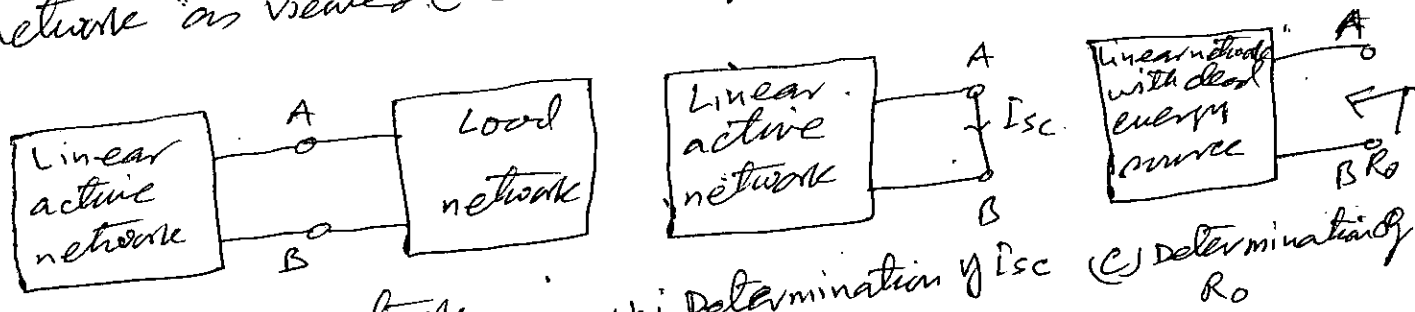
$$I = \frac{V\angle 0^\circ}{Z} = \frac{220\angle 0^\circ}{22.276\angle 80.96^\circ} = 9.87\angle -80.96^\circ\ A.$$

$$PF = \cos\phi = \cos(80.96) = 0.157 \text{ lagging.}$$

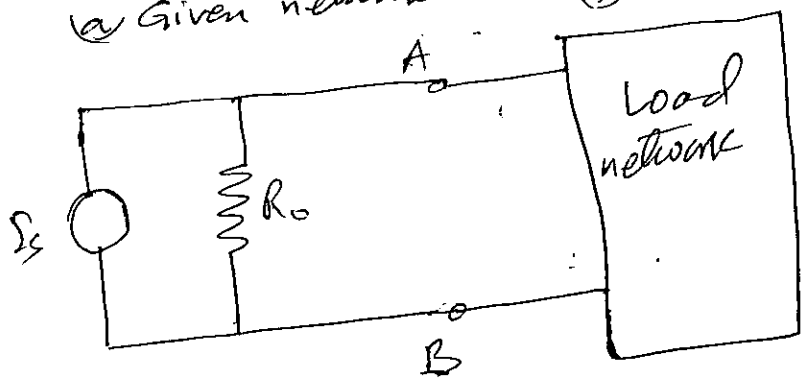


NORTON'S THEOREM

Any two terminals linear, active electrical network can be replaced by an ideal current source I_{sc} in parallel with a resistance R_0. where I_{sc} is the current which flows through the two terminals when they are short circuited and R_0 is the equivalent resistance of the network as viewed (measured from the terminals.

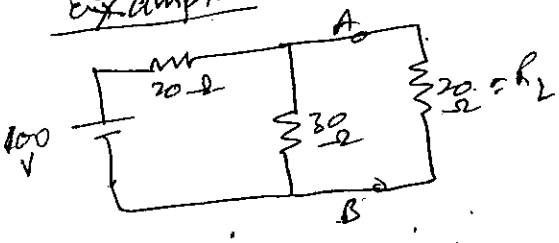


(a) Given network (b) Determination of I_{sc} (c) Determination of R_0

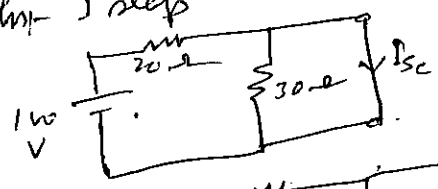


(d) Norton's Equivalent network.

Example:- Find the current in 20 Ω resistor.



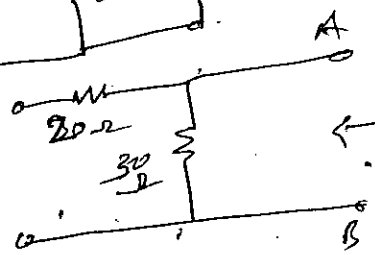
Soln:- I step



i step

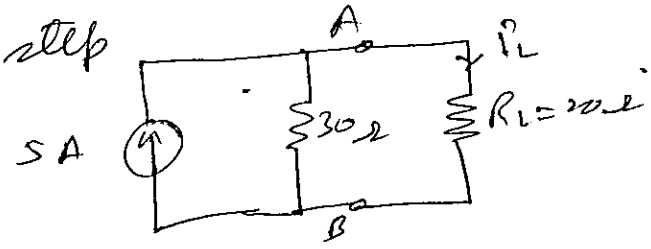
$I_{sc} = \frac{100}{20} = 5$ Amp.

ii step



$R_0 = 30\,\Omega$

iv step

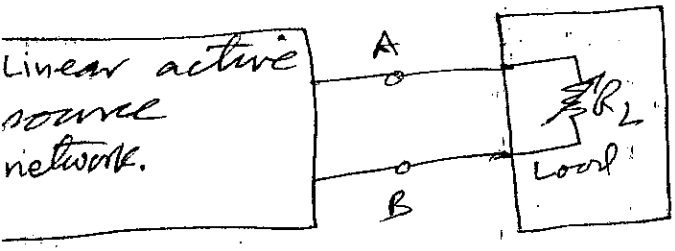


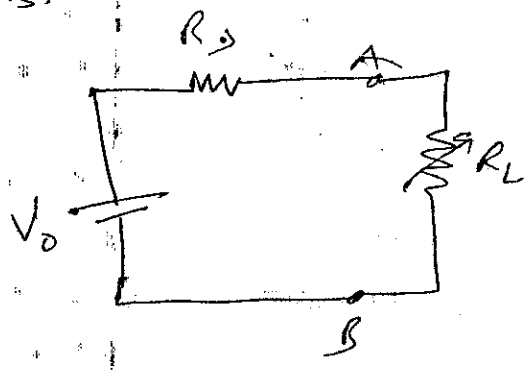
$\therefore I_L = \frac{30}{30+20} \times 5 = 3$ Amp

Ans.

Maximum Power Transfer Theorem

Maximum power will be transferred from a source network to load if the load resistance R_L is equal to the equivalent resistance of the source network R_s.





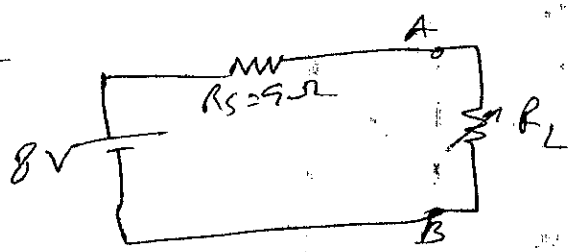
$$I_{max} = \frac{V_o}{R_s + R_L} = \frac{V_o}{2R_L}$$

$$P_{max} = I_{max}^2 \times R_L$$

$$= \frac{V_o^2}{4R_L^2} \times R_L = \frac{V_o^2}{4R_L}$$

$$\boxed{\therefore P_{max} = \frac{V_o^2}{4R_L}}$$

mple:



max^m power transfer $R_s = R_L$

$$\therefore P_{max} = \frac{V_o^2}{4R_L} = \frac{8^2}{4 \times 9} = \frac{64}{36} = 1.78 \text{ watts.}$$

an $R_L = 7\Omega$

$$= \frac{V_o}{R_s + R_L} = \frac{8}{9+7} = 0.5A \quad \therefore P = I^2 R_L = (0.5)^2 \times 7 = 1.75 \text{ watts.}$$

an $R_L = 11\Omega$

$$= \frac{V_o}{R_s + R_L} = \frac{8}{9+11} = 0.4A \quad \therefore P = I^2 R_L = (0.4)^2 \times 11 = 1.76 \text{ watts.}$$

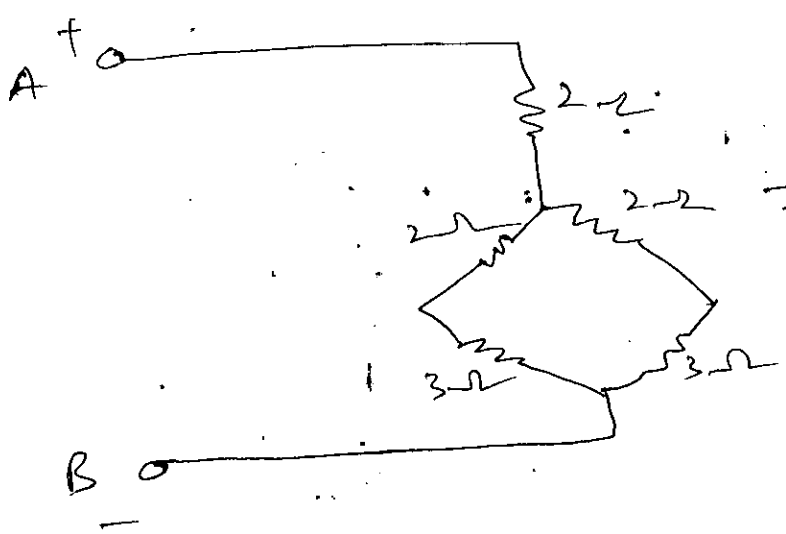
ence power transfer will be maximum when $R_L = R_s$

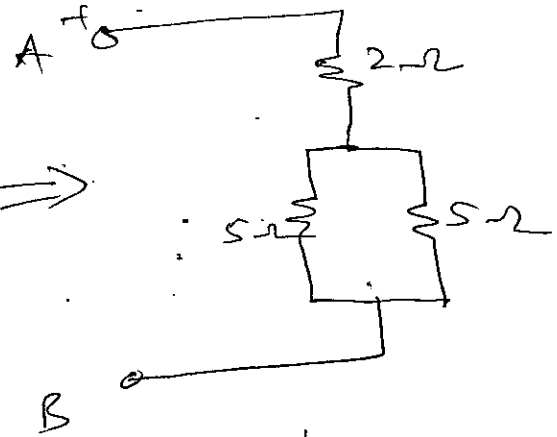
ہر سوال کا جواب ممکن ہے

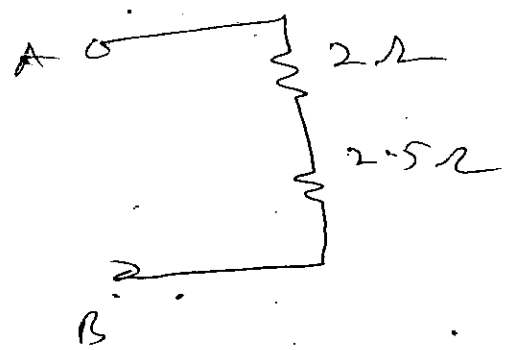
$R_a = \frac{6 \times 6}{6+6+6} = \frac{36}{18} = 2\,\Omega$, $R_b = \frac{6 \times 6}{6+6+6} = \frac{36}{18} = 2\,\Omega$

$R_c = \frac{6 \times 6}{6+6+6} = \frac{36}{18} = 2\,\Omega$.

Therefore new network becomes







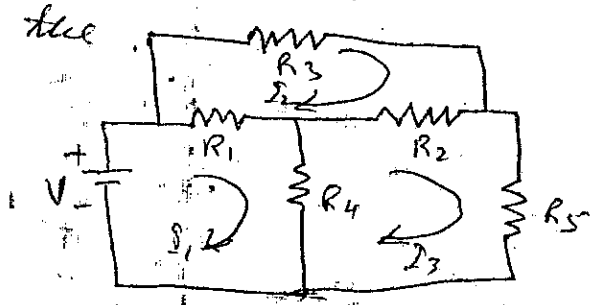
$\therefore R_{eq} = 2 + 2.5$

$= 4.5\,\Omega$ Ans

esh Analysis :- A mesh is defined as the allest closed path in a circuit. In any mesh age sum is zero, i.e sum of voltage drops & voltage rec is zero as given by Kirchhoff's voltage law. method which is based on Kirchhoff's voltage is known as Mesh Analysis. This is also led loop analysis.

cedure of mesh Analysis.

identify meshes existing in the rcuit. In the figure shown, e are three meshes.

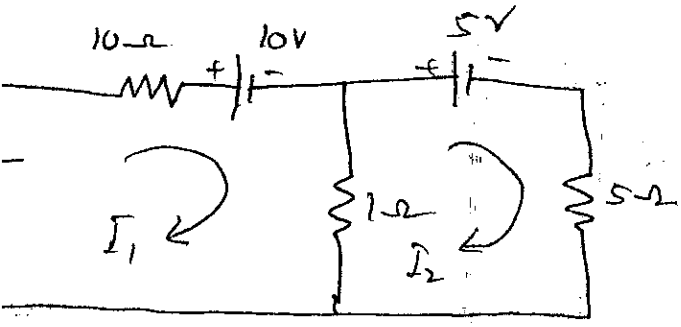


ssume mesh current as own I_1, I_2, I_3. (All in clockwise or Anticlockwise).

rite KVL equations for all loops and solve equations to find unknown quantities.

ere is any current source then convert it equivalent voltage source.

uple :- Find the current and power loss in 5Ω resistor ie fig. shown below.



Soln :- (a) There are two meshes 1 & 2.

(b) There are two currents I_1 & I_2 flowing in clockwise direction.

(c) Write KVL for both

hes. For mesh I, KVL equation will be

$-I_1 \times 10 - 10 - (I_1 - I_2) \times 1 = 0$ or $20 - 10I_1 - 10 - I_1 + I_2 = 0$

$11 I_1 - I_2 = 10$ —(1)

For mesh 2, KVL will be

$(I_1 - I_2)\times 1 - 5 - I_2 \times 5 = 0$

or $I_1 - I_2 - 5 - 5I_2 = 0$

or $I_1 - 6I_2 = 5$ —(2)

Multiplying eqn (2) by 11 & substracting from eqn (1), we have

$$\begin{array}{r} 11I_1 - I_2 = 10 \\ \underline{11I_1 - 66I_2 = 55} \\ \hline 65I_2 = -45 \end{array}$$

or $I_2 = -\frac{45}{65} = -\frac{9}{13}$ Amp. (This is current through 5 Ω resistor),

Substituting value of I_2 in eqn (1), we have

$11I_1 - \left(-\frac{9}{13}\right) = 10$

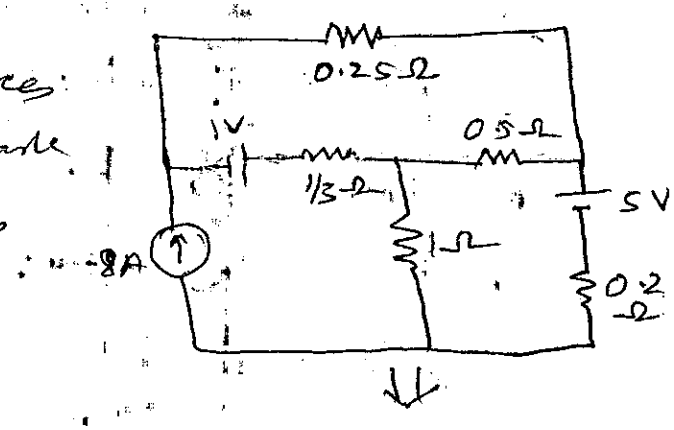
or $11I_1 + \frac{9}{13} = 10$ or $11I_1 = 10 - \frac{9}{13}$ or $11I_1 = \frac{121}{13}$ or $I_1 = \frac{11}{13}$

Power loss in 5 Ω resistor

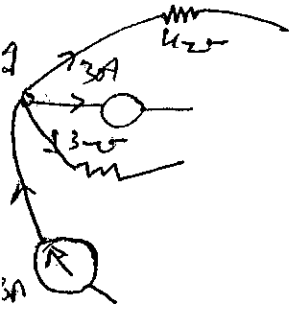
$P = I^2 R = \left(\frac{-9}{13}\right)^2 \times 5 = \frac{81}{169} \times 5 = 2.4$ watt Ans

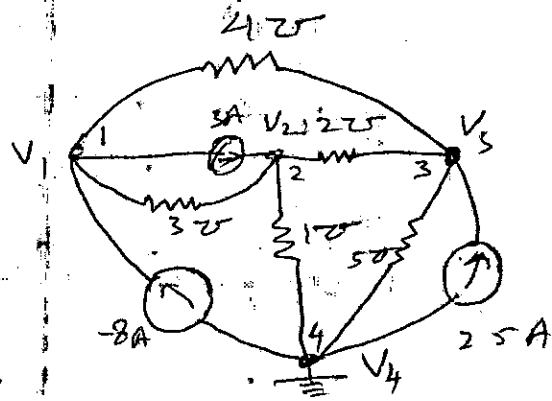
<u>NOTE</u> :- If in any mesh, a current source can not be converted into voltage sources and this current source is not common in two meshes, then no need to write the KVL for this mesh, because current is already known.

odal Analysis:- Nodal analysis is more powerful
nique to solve electric circuits than mesh analysis.
odal analysis, we consider nodes which are connected
t least three elements. The nodal analysis is
) on KCL. In any circuit first point out no. of
, which are connected to atleast three elements.
ecedure:- (a) First of all mark nodes as a, b, c, ---.
lect one node as reference node.
ltage of reference node is always taken as zero.
rite voltage of remaining nodes as V_a, V_b, -- etc.
rite KCL for every node except reference node &
e KCL equations to find unknown variables.
nvert voltage source into current source.
ple:- Solve the circuit shown below using nodal
ysis.



- (a) convert voltage sources
current sources and mark
s. Also convert resistances
conductances.
rite KCL at node 1.



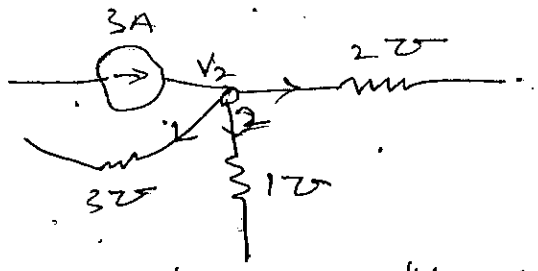


1 has four branches connected
own above. Voltage of node under
eration is always taken as higher
other nodes. Hence current flows from the node.
at node 1 will be.

$(V_1-V_2)\times 3 + (V_1-V_3)\times 4 + 3 - (-8) = 0$ or $3V_1 - 3V_2 + 4V_1 - 4V_3 + 3 + 8 = 0$

or $7V_1 - 3V_2 - 4V_3 = -11$ —(

(c) Write KCL at node 2.

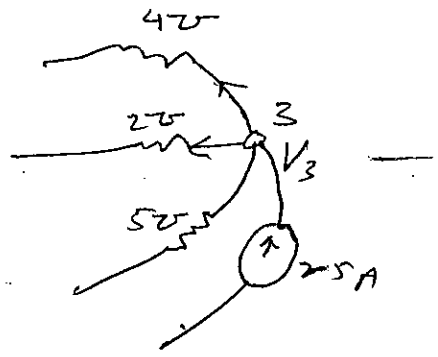


$(V_2-V_1)\times 3 + (V_2-V_4)\times 1 + (V_2-V_3)\times 2 - 3 = 0$

or $3V_2 - 3V_1 + V_2 - V_4 + 2V_2 - 2V_3 = 3$

$-3V_1 + 6V_2 - 2V_3 = 3$ — ② since $V_4 = 0$

(d) Write KCL for node 3.



$(V_3-V_1)\times 4 + (V_3-V_2)\times 2 + V_3\times 5 = 25$

or $4V_3 - 4V_1 + 2V_3 - 2V_2 + 5V_3 = 25$

or $-4V_1 - 2V_2 + 11V_3 = 25$ — ③

In matrix form eqns ①, ② & ③ can be written as

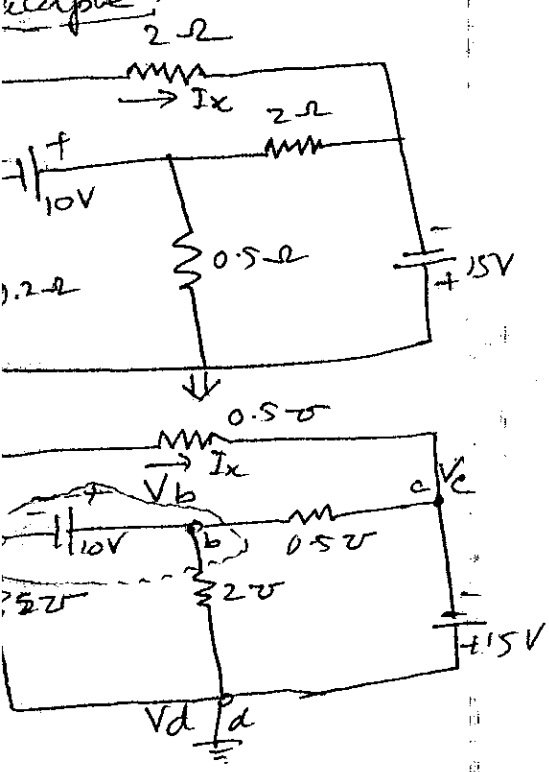
$$\begin{bmatrix} 7 & -3 & -4 \\ -3 & 6 & -2 \\ -4 & -2 & 11 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} -11 \\ 3 \\ 25 \end{bmatrix}$$

$$\therefore V_1 = \frac{\begin{bmatrix} -11 & -3 & -4 \\ 3 & 6 & -2 \\ 25 & -2 & 11 \end{bmatrix}}{\begin{bmatrix} 7 & -3 & -4 \\ -3 & 6 & -2 \\ -4 & -2 & 11 \end{bmatrix}} = \frac{9}{9} = 1 \text{ Volts}$$

Similarly $V_2 = 2$ Volts & $V_3 = 3$ Volts Ans.

:- If a voltage source connected in the network
y be converted into a current source, then such
tage source is called unaccompanied voltage
:. If an unaccompanied voltage source is connected
n reference node and any other node, then for
node, no need to write nodal equation, because
tage of this node is already given.
If unaccompanied voltage source is connected
een any two nodes other than reference node
in such a case a supernode will have to
made and for this supernode, KCL will be
cited to write.

xample :- Find the current I_x in the circuit shown.



Soln :- (a) Identify nodes and one reference node as a, b, c, d
(b) Mention node voltages as $V_a, V_b, V_c, \ldots$ etc.
(c) convert resistance into conductance.

Note :- There are two voltage sources, both are unaccompanied voltage sources, because they can not be converted into equivalent current sources.

oltage source between nodes a & b is not connected to
eference node & hence a & b will make supernode.

KCL for super node is

$V_a \times 2 + (V_a - V_c) \times 0.5 + (V_b - V_c) \times 0.5 + V_b \times 2 = 0$

$2V_a + 0.5V_a - 0.5V_c + 0.5V_b - 0.5V_c + 2V_b = 0$

$2.5V_a + 2.5V_b - V_c = 0$ — ①

Also supernode gives one KVL, i.e.

$V_a + 10 = V_b$ — ② or $V_a - V_b = -10$ — ②

No need to write KCL for node c, because $V_c = -15$ volt given and $V_d = 0$

Substituting $V_c = -15$ in eqn ①, we have,

$5.5V_a + 2.5V_b - (-15) = 0$ or $5.5V_a + 2.5V_b = -15$ — ③

or $V_a + V_b = -6$ — ③

Adding eqn ② & ③, we have

$V_a - V_b = -10$ — ②

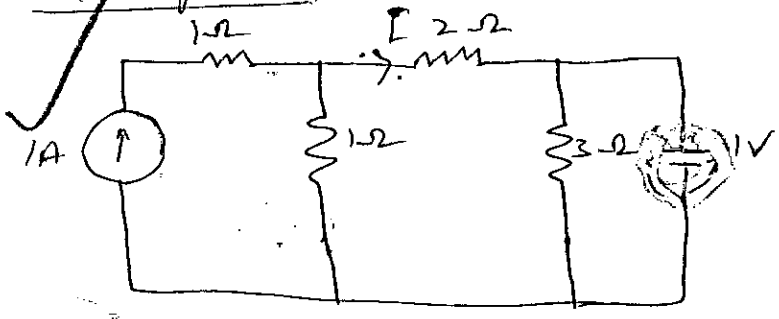
$V_a + V_b = -6$ — ③

$2V_a = -16$

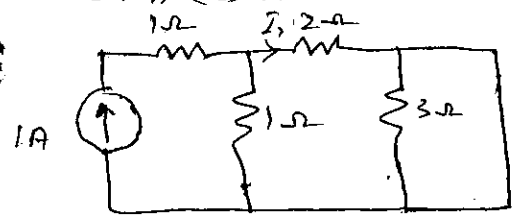
or $V_a = -8$ volt

$I_{2\Omega} = \frac{(V_a - V_c)}{2} 0.5 = [-8 - (-15)] \times 0.5 = 7 \times 0.5 = 3.5$ ~~volt~~ Amp Ans

Example-2: Find current I in 2Ω resistor.



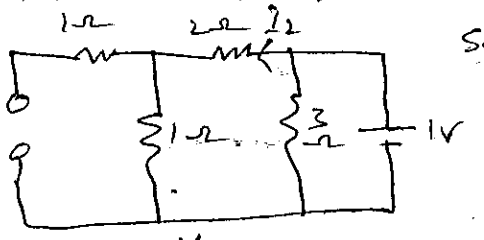
Soln:- (a) First consider 1A current source only.



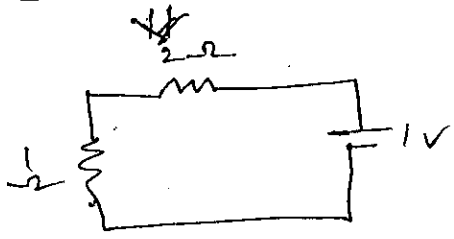
Suppose current I_1 is flowing in 2Ω resistor due to current source only,

$I_1 = 1 \times \frac{1}{1+2} = \frac{1}{3}$ Amp

(b) Now consider 1V voltage source only.



Suppose I_2 current is flowing in 2Ω resistor due to voltage source only.



$I_2 = \frac{V}{R} = \frac{1}{1+2} = \frac{1}{3}$

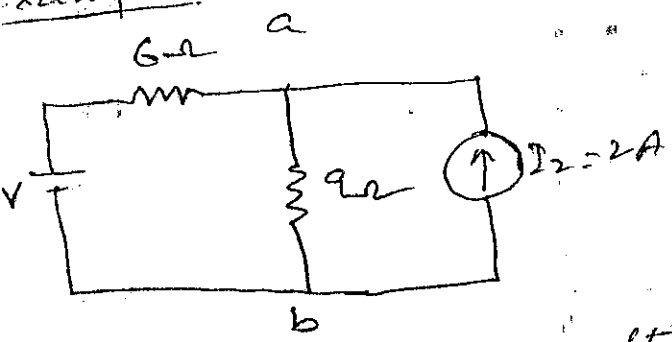
Net current $= I = I_1 - I_2$

$= \frac{1}{3} - \frac{1}{3} = 0$ Ans

Superposition Theorem:-

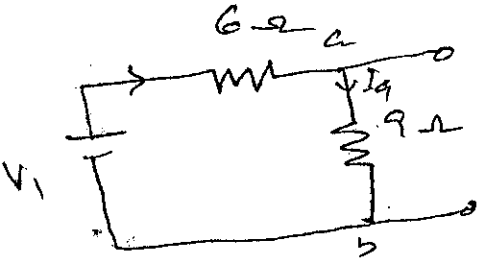
It states that in any electrical network having ear elements and more than one independent urce, the current through any branch or voltage ross any branch is equal to algebraic sum ce to one source at a time. i.e consider only e source at a time and make all other sources ual to zero.

xample1:- Find current in 9Ω resistance.

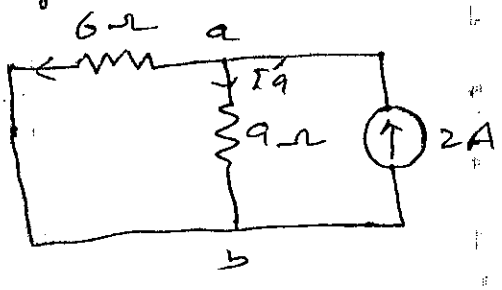


, let us first consider voltage source V_1. To make current source zero, remove it.

$$I_9 = \frac{V_1}{6+9} = \frac{3}{15} = 0.2 \text{ Amp.}$$



(b) Now consider only current source I_2. To make voltage source V_1 zero. Replace V_1 by short circuit.



$$I'_9 = 2 \times \frac{6}{6+9} = \frac{12}{15} = 0.8 \text{ Amp.}$$

Hence total current in 9Ω resistor $= I_9 + I'_9$

$= 0.2 + 0.8$

$= 1$ Amp.

Ans

UNIT-4

ROTATING ELECTRICAL MACHINES

D.C. Generator

An electrical generator is a machine which converts mechanical energy (or power) into electrical energy (or power).

This energy conversion is based on Faraday's law of electromagnetic induction. This law states that whenever a conductor cuts a magnetic flux or magnetic flux cuts conductor, an e.m.f is induced across the conductor. This induced e.m.f causes a current to flow if the conductor circuit is closed.

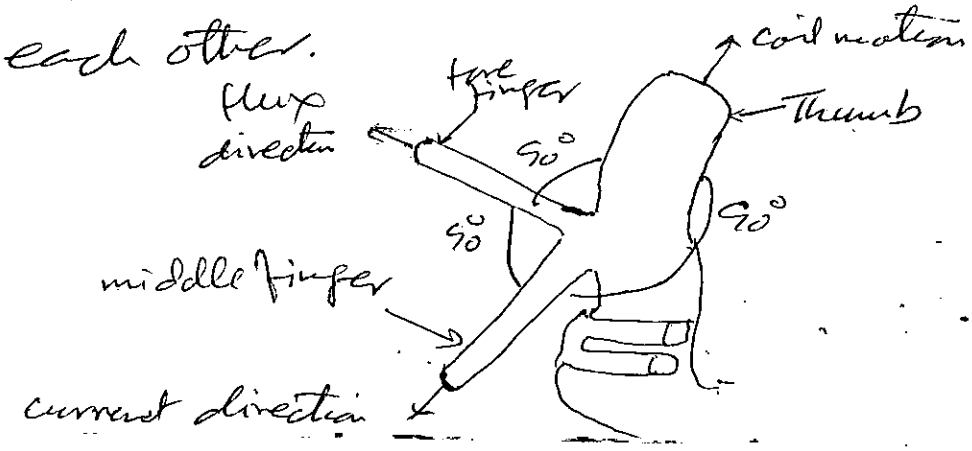
Hence the essential parts of an electrical generator are

(i) A magnetic field.

and (ii) Conductor (or coil).

The direction of flux, current produced and motion of coil is given by Flemings rule.

(i) Flemings right hand rule is applied to generators. According to this rule, fore finger gives direction of flux, middle finger gives direction of current & thumb gives direction of motion of coil. The thumb, fore finger & middle finger should be at right angles to each other.



Flemings left hand rule is applied to D.C motors. in this case also fore finger gives direction of flux, middle finger gives direction of current thumb gives direction of motion of conductor.

Essential parts of a D.C. Generator.

Magnetic frame or Yoke:

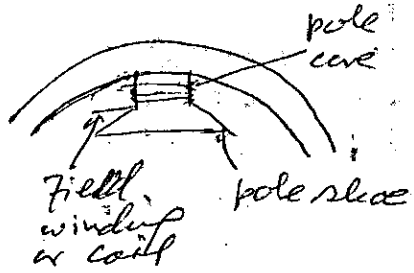
The outer frame or yoke serves double purpose as below.

- provides mechanical support for poles and also it is a cover of whole machine.
- carries magnetic flux produced by poles.

Pole Cores and Pole shoes.

The magnets which create magnetic flux consists of core, pole shoe and field winding.

cross sectional area of pole shoe larger than pole core. This is to hold field coil or field winding. large area of pole shoe spread with low reluctance of magnetic path.



Pole coil or field coil:

As shown in the fig., coil placed on pole core called pole coil or field coil or winding. This create magnetic flux.

Armature core:- Armature core is made up of laminations. Armature core is cylindrical in shape and is mounted on shaft.

(v) Armature winding :- Armature winding or armature coil is placed at inter surface of armature core. Armature coil carries current produced by D.C generator.

(vi) Commutator :- In DC generator, AC current is generated. To make AC into DC, commutator is used.

(vii) Bushes :- Bushes are placed on commutator to collect current.

In case of DC generator, rotor is rotated by a turbine and hence magnetic flux produced by field magnets cut and emf is induced in the coil.

EMF equation is as follows.

$$EMF, E = \frac{\phi Z N P}{60 A} \text{ volts}$$

where ϕ = flux in Wb
Z = No. of conductors
N = Rotation of armature coil in r.p.m.
P = No. of poles
A = No. of parallel paths.

The D.C generator can be used as a D.C motor also. If we rotate the rotor, the machine works as a DC generator. If we give electric supply to the rotor, the machine works as a DC motor.

Application of D.C. machines.

D.C motors are used in lathes, drills, spinning and
aving machines, elevators & traction, cranes, elevator
compressor etc.

INDUCTION MOTORS

Among all type of A.C. motors, induction motors are widely used for various purposes. It has no. of advantages and few disadvantages also.

Advantages of Induction motors.

(a) It has very simple and almost unbreakable construction
(b) Its cost is low and it is very reliable.
(c) Its efficiency is very high.
(d) Its power factor is very good.
(e) It requires minimum of maintenance.
(f) Its starting arrangement is simple.

Disadvantages of Induction motors.

(a) By varying its speed, its efficiency decreases.
(b) Its speed decreases with increase in load.
(c) Its starting torque is low as compared to D.C shunt motor.

Application of Induction motors.

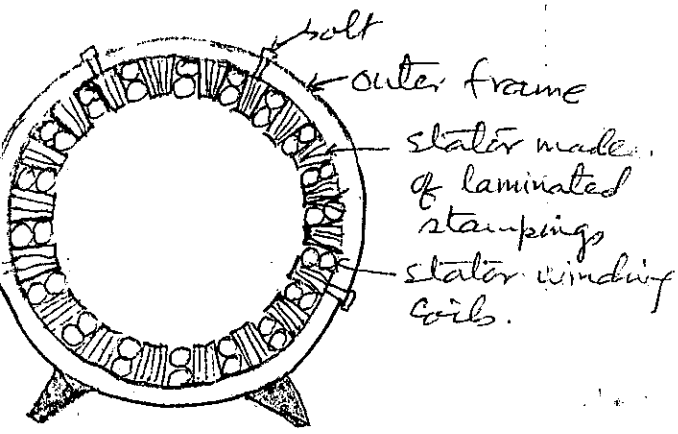
widely used in almost all domestic, industrial and agricultural purpose.

Construction of Induction Motors.

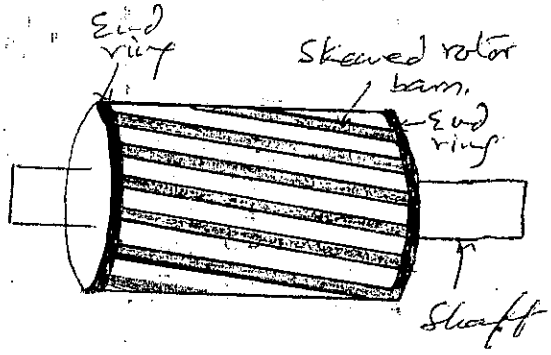
An induction motor mainly consists of two main arts, namely

i) A Stator (b) A rotor.

Stator :- The stator of an induction motor is made of thin laminations of high quality, low loss silicon el. These laminations have slots at inner side in ich stator winding is fitted. outer side of stampings bolted to frame or cover of induction motor. outer ve is made up of cast iron.



Stator



Rotor

Just like stator, rotor of an induction motor also made up of thin laminations of high quality loss silicon steel. Because low loss silicon steel inations provide low reluctance path to the main x. There are two type of rotors.

Squirrel cage rotor :- About 90% of induction motors constructed with squirrel cage type rotor because is very simple and most rugged construction. rotor consists of cylindrical laminated core having illel slots to receive rotor bars as shown in fig. ore. These rotor bars are welded at two ends

with the help of end rings.

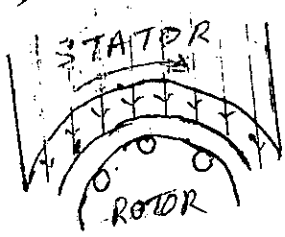
The advantage of squirrel cage rotor is that it is very simple and rugged in construction. The disadvantage is that its starting torque can not be increased.

(b) <u>Phase-wound Rotor</u> :- This type of rotor is provided with 3-phase, double layer, distributed winding. The number of rotor poles are equal to the number of stator poles. The one end of all the three phases are connected in star and another end of all the three phases of the rotor winding are connected to slip rings separately. These slip rings are insulated from each other and mounted on shaft. Three brushes are rested on these three slip rings. The brushes are connected to three phase rheostat to increase the resistance of the winding at the time of starting. By increasing resistance of the rotor at starting time, rotor torque can be increased. When rotor catch up its speed, the brushes are automatically lifted and slip rings are short circuited, hence making closed circuit.

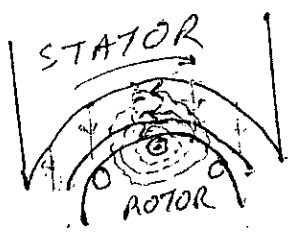
Why does the rotor of an Induction Motor rotates?

When the stator winding of an induction motor is given a 3-phase (or two phase) supply, then a constant flux sets up which revolves at synchronous speed ($N_s = 120f/P$). This revolving flux cuts the rotor conductors. The rotor is yet stationary. Due to flux cutting an emf is induced in the rotor bars. Since rotor circuit is closed and hence current is induced in the rotor circuit. The direction of this induced current is in opposition to the cause which produces the current. The current is produced due to relative speed between revolving flux and rotor. Hence due to opposition of induced current, rotor starts rotating.

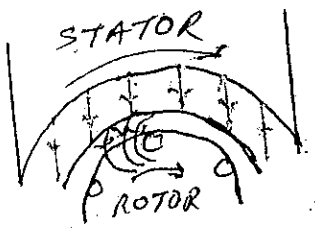
Another reason of revolving rotor is that, the constant flux produces current in rotor winding and due to interaction of main flux and current a force is exerted on each rotor winding as per Flemings Left Hand rule. Due to this force, rotor starts rotating.



STATOR FLUX ONLY



ROTOR FLUX ONLY



COMBINED FLUX

Synchronous Generator.

Synchronous generator or A.C generator or alternator is same machine. Synchronous generator also works on the principle of electromagnetic induction. As we have studied in D.C generators, synchronous generators also have two windings namely.

(a) Field winding.

(b) Armature winding.

In D.C generators, field winding is provided on stator and armature winding is provided on rotor.

In case of synchronous generators, the system is opposite. Here field winding is provided on rotor and armature winding is provided on stator.

Therefore in case of synchronous generator, armature is stationary.

EMF is induced in armature winding and hence generated current is generated in the armature winding.

The advantages of stationary armature winding are as follows.

(1) Since armature is stationary, generated current can be taken out directly by armature terminals.

(2) It is easy to insulate stationary winding

(3) Since armature winding is stationary, hence no any damage chance.

(4) The sliding contacts, i.e slip rings are transferred to low voltage, low power d.c field supply.

Principle of operation of synchronous generator.

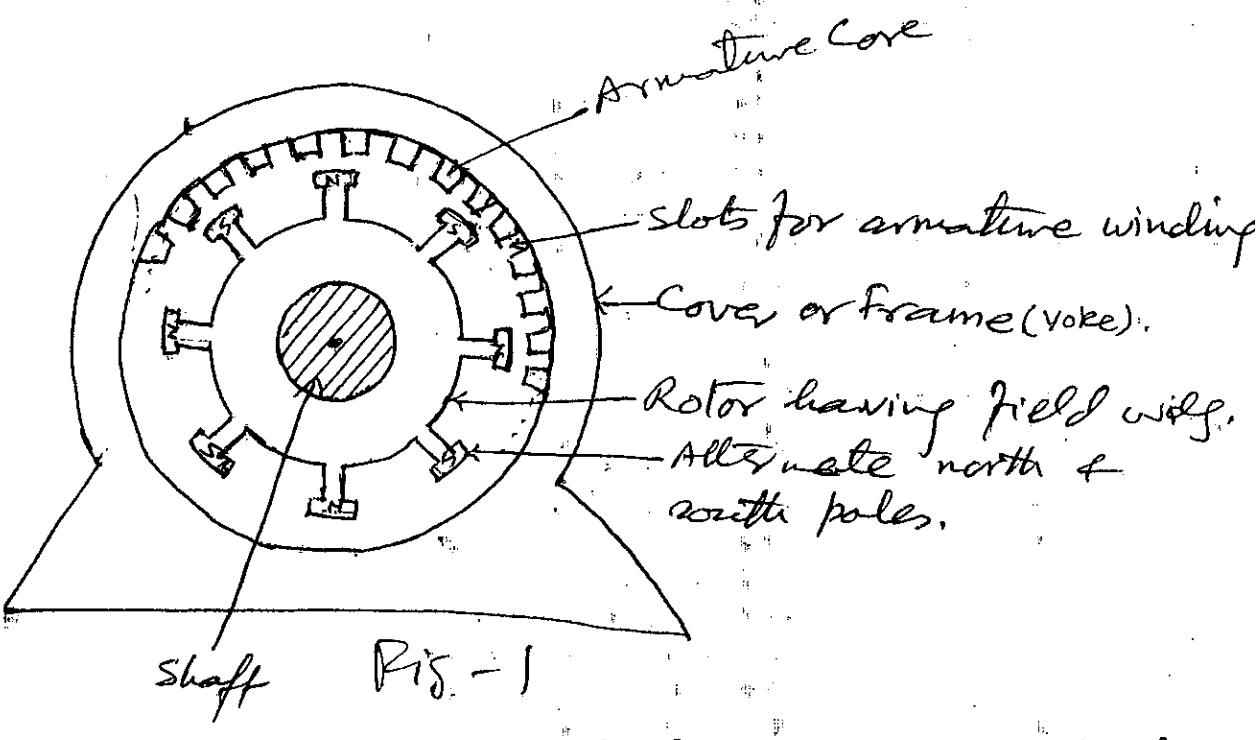


Fig. - 1

When the rotor is rotated at a particular speed, then alternate north & south poles mounted on the rotor also rotates. These rotating magnetic poles distribute rotating flux which cuts the armature windings in the stator. Due to this cut of magnetic flux, e.m.f is induced in the armature winding and hence a current is produced which is taken out through leads.

Consider an ideal transformer [Fig. 27-13 (*a*)] whose secondary is open and whose primary is connected to sinusoidal alternating voltage V_1. This potential difference causes an alternating

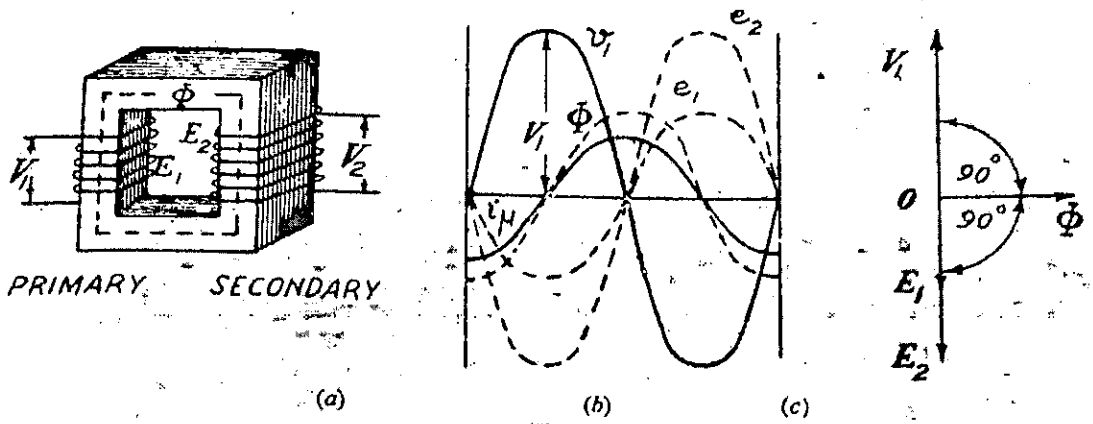


Fig. 27-13

current to flow in primary. Since the primary coil is purely inductive and there is no output (secondary being open) the primary draws the magnetising current I_M only. The function of this current is merely to magnetise the core, it is small in magnitude and lags V_1 by 90°. This alternating current I_M produces an alternating flux Φ which is, at all times, proportional to the current (assuming permeability of the magnetic circuit to be constant) and, hence, is in phase with it. This changing flux is linked both with the primary and the secondary windings. Therefore, it produces self-induced e.m.f. in the primary. This *self-induced* e.m.f. E_1 is, at every instant, equal to and in opposition to V_1. It is also known as *counter* e.m.f. or back e.m.f. of the primary.

Similarly, there is produced in the secondary an induced e.m.f. E_2 which is known as *mutually* induced e.m.f. This e.m.f. is antiphase with V_1 and its magnitude is proportional to the rate of change of flux and the number of secondary turns.

The instantaneous values of applied voltage, induced e.m.f.s, flux and magnetising current are shown by sinusoidal waves in Fig. 27-13 (*b*). Fig. 27-13 (*c*) shows the vectorial representation of the effective values of the above quantities.

27-6. E.M.F. Equation of a Transformer

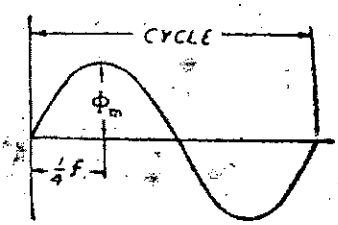


Fig. 27-14

Let N_1 = No. of turns in primary
N_2 = No. of turns in secondary
Φ_m = Maximum flux in core in webers $= B_m \times A$
f = frequency of a.c. input in Hz

As shown in Fig. 27-14, flux increases from its zero value to maximum value Φ_m in one quarter of the cycle *i.e.* in $1/4f$ second

$$\therefore \quad \text{Average rate of change of flux} = \frac{\Phi_m}{1/4f} = 4f\Phi_m \text{ Wb/s or volt}$$

Now, rate of change of flux per turn means induced e.m.f. in volts.

$\therefore$ Average e.m.f./turn $= 4f\Phi_m$ volt

If flux Φ varies *sinusoidally*, then r.m.s. value of induced e.m.f. is obtained by multiplying the average value with form factor.

$$\text{Form factor} = \frac{\text{r.m.s. value}}{\text{average value}} = 1{\cdot}11$$

$\therefore$ r.m.s. value of e.m.f./turn $= 1{\cdot}11 \times 4f\,\Phi_m = 4{\cdot}44\,f\,\Phi_m$ volt

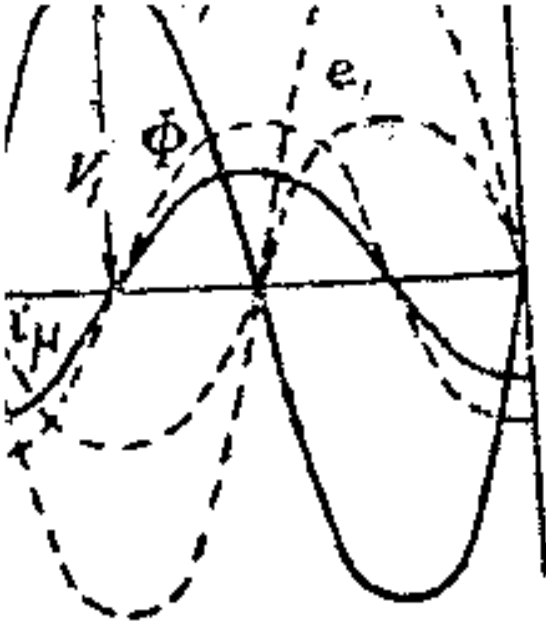
Now, r.m.s. value of induced e.m.f. in the whole of primary winding

$= (\text{induced e.m.f./turn}) \times \text{No. of primary turns}$

$$\therefore \quad E_1 = 4{\cdot}44\,f\,N_1\Phi_m = 4{\cdot}44\,fN_1B_m\,A \qquad \ldots(i)$$

Similarly, r.m.s. value of e.m.f. induced in secondary is,

$$E_2 = 4{\cdot}44\,fN_2\Phi_m = 4{\cdot}44\,fN_2B_m\,A \qquad \ldots(ii)$$



(b) (c

Fig. 27-13

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